

Nine-Species MEA ePC-SAFT Parameter Bundle

Update-in-place research notebook

3 September 2026

Status: active MEA parameter bundle. This repository owns the live MEA bundle recorded here. It uses solvent-only mass-fraction dielectric mixing, automatically activated extended Born (SSM+DS), the displayed R4 carbamate-hydrolysis correlation, ϵ_{CO_2}/k , and R5 $a_K = 2597.91$ K. The exact active replay evaluates 161/161 pressure states with $\log_{10}$ RMSE 0.3317 and 125/131 positive observed speciation targets with $\log_{10}$ RMSE 0.3702. Value color alone identifies one of the four evidence classes; parameter values use regular weight so typography does not imply a second status system. This PDF is the live present-state view: new accepted evidence replaces the corresponding prior interpretation in place, while Git preserves history.

1. Question and scope

What full parameter bundle is being used for the current calculations? This notebook records the nine-species model structure, pure-component values, association graph, sparse binary interactions, electrostatic choices, reaction correlations, and known non-adopted diagnostics in one reviewable view. Section 9 replays the exact active vector after the audited R1–R3 standard-state correction using its pinned Engine wheel; Section 8 shows its current direct-Engine speciation grid. The reported results use the parameter set shown in Tables 1–4.

The liquid species are CO_2 , MEA, H_2O , MEA^+H , MEACOO^- , HCO_3^- , CO_3^{2-} , H_3O^+ , and OH^- . The neutral vapor species and standard-state details remain governed by the future immutable packet.

1.1. How to read this live notebook

verified statements are checked against the retained input or result files, the current repository, an official software source, or a located primary source. **inference** statements interpret those checked facts. **assumption** identifies a proposed test condition, and **unknown** marks a question that the retained evidence does not answer. Unrun experiments are listed only in Section 12; they are not results. This working notebook has not been independently reviewed or promoted for manuscript use.

Evidence	Present state	Meaning
verified	Parameter set	Tables 1–4 contain the active nine-species vector retained here.
verified	Current replay	The automatic-SSM+DS replay evaluates 161/161 pressure states and 42/44 observed speciation states (125/131 positive targets). The display grid evaluates 181/184 states and retains its three failed points explicitly. Current fixed-structure metrics and every failed attempt are reported in Section 11.2.
verified	Upstream work	ePC-SAFT Issues #80, #119, #126, and #149 remain open. The native Regression and repository-owner cutovers have merged through upstream PR #169, so Issue #119's old pre-build blocker text is stale even though the scientific campaign has not run.

2. Model conditions

Choice	Current foundation
Residual Helmholtz terms	PC-SAFT hard-chain and dispersion; general-site association; Debye–Hückel and Born electrostatics.
Association topology	Reciprocal induced CO ₂ -H ₂ O cross association plus ordinary MEA and water association. No CO ₂ self-association and no separate CO ₂ -MEA chemical-association edge.
Permittivity and Born formulation	Solvent-only mass-fraction mixing: normalized MEA–water mass fractions determine bulk ϵ_r ; dissolved CO ₂ and ions do not enter the mixing rule. Populated unique Born diameters and the non-unit water solvation factor automatically select corrected SSM+DS; no independent on/off switches are stored.
Parameter identity	Active solvent-only-mass-fraction/automatic-SSM+DS bundle with the selected R5 adjustment. Parameter fingerprint: sha256:163ea97a97d3a6ee0c135ed0cad891fd33689ca135c53e694097fad1c21e4788.
Refinement boundary	The selected bundle changes R4, R5 a_K , and ϵ_{CO_2}/k from the literature-initialized foundation. R1–R3, the remaining R5 coefficients, ion block, association topology, and binary interactions remain fixed.

3. Complete component parameter bundle

Here m is segment number, σ is segment diameter, ϵ/k is dispersion energy, ϵ^{AB}/k is self-association energy, and κ^{AB} is the stored site-volume coordinate. For CO₂, zero self-association energy disables self-association and the listed site volume is used only by the induced cross-association set. Here z is charge and d_b is Born diameter, ϵ_r is relative permittivity, and f_{solv} is the solvation factor. Lengths are in Å and energies in K. References identify the literature lineage; the underlying exact vector is results/selected-current-best-parameters.json. The Debye–Hückel diameter is derived as $d_{\text{DH},i} = 0.88\sigma_i$ and is therefore not tabulated as an independent parameter. The hash in Section 2 identifies the unrounded authority behind the four-decimal-place display.

Table 1. Neutral-component parameters. Only parameters applicable to the neutral molecular rows are shown.

Species	Scheme	m_i	σ_i	ϵ_i/k	ϵ_i^{AB}/k	κ_i^{AB}	$\epsilon_{r,i}$	$f_{\text{solv},i}$	Ref.
CO ₂	2B ind.	2.0729	2.7852	173.4403	0.0000	0.0451	— ^a	1.0000	[8, 9, 16, 18]
MEA	2B	3.0353	3.0435	277.1740	2586.3000	0.0375	32.0000	1.0000	[1, 2, 4, 8]
H ₂ O	2B	1.2047	$\sigma(T)$	353.9500	2425.7000	0.0451	$\epsilon_r(T)$	1.5000	[8, 11, 17, 18]

^a Go to Section 5.2 for the source, exact derivation, temperature dependence, and replacement decision. The f_{solv} values are active in the selected automatic SSM+DS formulation.

Table 2. Ionic-species parameters. Only parameters applicable to the ionic rows are shown.

Species	σ_i	ϵ_i/k	z_i	$\epsilon_{r,i}$	$d_{B,i}$	Ref.
MEAH ⁺	3.4851	232.6872	+1	8.0000	3.5332	[8, 19, 20]
MEACOO [−]	3.5354	453.2652	−1	8.0000	3.5411	[8, 19, 20]
HCO ₃ [−]	2.9296	70.0000	−1	8.0000	3.0000	[8, 12, 18]
CO ₃ ^{2−}	2.4422	249.2600	−2	8.0000	3.0000	[8, 12, 18]
H ₃ O ⁺	3.4654	500.0000	+1	8.0000	1.2180	[8, 12]
OH [−]	2.0177	650.0000	−1	8.0000	3.0811	[8, 12]

The invariant conventions $m_i = 1$ and $f_{\text{solv},i} = 1$ apply to every ionic species and are omitted as non-discriminating columns. The fixed $\epsilon_{r,i} = 8$ convention applies to every ion whenever a formulation requires component-level ionic permittivity.

Table 3. Binary interaction matrix, k_{ij} . Upper triangle shown; the diagonal is structural and unclassified.

Species	CO ₂	MEA	H ₂ O	MEAH ⁺	MEACOO [−]	HCO ₃ [−]	CO ₃ ^{2−}	H ₃ O ⁺	OH [−]
CO ₂	—	0.0000	0.0133	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MEA		—	−0.0735	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
H ₂ O			—	0.0000	0.0000	0.0000	−0.2500	0.2500	−0.2500
MEAH ⁺				—	−0.0020	0.0000	0.0000	1.0000	0.0000
MEACOO [−]					—	1.0000	1.0000	0.0000	1.0000
HCO ₃ [−]						—	1.0000	0.0000	1.0000
CO ₃ ^{2−}							—	0.0000	1.0000
H ₃ O ⁺								—	0.0000
OH [−]									—

Class 1: fixed; no active refinement indicated

Class 3: local value or demonstrated refinement candidate

Class 2: supported value with a specific comparison still needed

Class 4: missing value, default, or indirect analog

4. Reaction parameter block

For R1–R3, the Austgen source form is $\ln K_{r,\text{source}} = A + B/T + C \ln T + DT$. Before constructing an Engine problem, the replay evaluates

$$\ln K_{r,\text{common}}(T) = \ln K_{r,\text{source}}(T) + \Delta_r, \quad r \in \{\text{R1}, \text{R2}, \text{R3}\}, \quad (1)$$

with $(\Delta_1, \Delta_2, \Delta_3) = (8.0330699846, 4.0165349923, 4.0165349923)$. These audited shifts convert the Austgen mixed mole-fraction reference to aqueous-molality-infinite-dilution-water-v1. R4 and R5 use their declared common-molality forms.

Table 4. Reaction-equilibrium parameters.

ID	A	B	C	D	Domain (K)	Definition
R1	132.899	−13445.9	−22.4773	0	273.15–498.15	Water autoprotolysis.
R2	231.465	−12092.1	−36.7816	0	273.15–498.15	CO ₂ hydration to bicarbonate.
R3	216.049	−12431.7	−35.4819	0	273.15–498.15	Bicarbonate dissociation.
R4	$\ln K = 3.3515778 - 1895.3/T$				293.15–393.15 ^a	Carbamate hydrolysis; selected fit.
R5	$\ln K = -\ln(10)[2597.91/T + 0.3869 + 0.0004277T]$				273.15–323.15	MEA ⁺ dissociation; selected a_K .

^aThe source correlation was qualified over 293.15–323.15 K. The displayed fit evaluates it through 393.15 K. R2 uses Austgen’s printed $A = 231.465$; the later 231.456 transcription is retained only as a documented source discrepancy.

5. Evidence-class justification

The class is assigned to each independent displayed value, not to an entire species. It reports the present refinement decision. Class 1 is fixed because the value is source-supported, a required model convention, inactive in the selected formulation, or not indicated for refinement by the retained comparisons. Class 2 is a supported active value with one specific transfer or sensitivity comparison still needed. Class 3 is a locally selected, fitted, or derived value, or a value already identified as a material refinement candidate. Class 4 is a missing independent value represented only by an unscreened default or indirect analog. Provenance and sensitivity remain stated separately so a low-priority parameter is not colored as uncertain merely because it came from literature.

5.1. Class 1: fixed current foundation

CO₂. The 2B induced scheme, $m = 2.0729$, and $\sigma = 2.7852 \text{ \AA}$ are the Gross–Sadowski molecular row carried through the induced-association literature. Zero self-association energy and $\kappa^{AB} = 0.0451$ implement the Pabsch–Schick induced topology rather than a fitted CO₂ self-association term [9, 16, 18]. The dispersion energy is no longer Class 1; its promoted local value is documented under Class 3.

MEA. The displayed m , σ , ϵ/k , ϵ^{AB}/k , and κ^{AB} are the Baygi 2B row [4]. The 2B topology is the current selection recorded by upstream PR 132; the older 3B diagnostic is not the retained foundation.

H₂O. The Held 2B family fixes m , $\sigma(T)$, ϵ/k , ϵ^{AB}/k , and κ^{AB} [11, 18]. Figiel et al. directly fit $f_{\text{solv},\text{H}_2\text{O}} = 1.5$ to NaBr mean-ionic-activity-coefficient data [8, Eqns. 8–10 and Tables 2–3].

Neutral-solvent permittivities. The retained water correlation is the Archer–Wang approximation and gives $\epsilon_{r,\text{H}_2\text{O}}(298.15 \text{ K}) = 78.4104$. This is close to, but not identical with, the high-accuracy 0.1-MPa value 78.3752 from Pátek et al.; the difference 0.0352 exceeds that reference’s stated uncertainty of about 0.01 [17, Eq. 11 and Tables 7–8]. Pátek’s correlation ends at 383.15 K, whereas this diagnostic reaches 393.15 K, so it cannot validate the upper endpoint. The retained $\epsilon_{r,\text{MEA}} = 32.0000$ is also a confirmed current foundation: Afify and Sweatman report an experimental literature average of 32.8 ± 2.1 at 298 K, and the direct dielectric-relaxation measurements of Agieienko and Buchner give 30.45 for neat MEA at 298.15 K [1, Table II] [2, p. 2314]. The displayed 32.0000 is the retained rounded working representative within that evidence range, not an exact measured value or temperature correlation and not a value reported by Baygi and Pahlavanzadeh. Baygi supports the other MEA PC-SAFT coordinates in Table 1 but contains no dielectric measurement or permittivity correlation.

All six ions. One segment, the signed integer charge, $\epsilon_{r,i} = 8$, and $f_{\text{solv},i} = 1$ are common model conventions rather than independently fitted single-ion properties. Figiel et al. use the same local ionic-region permittivity for every ion and set every pure-ion solvation factor to one because a hypothetical pure-ion system contains no solvent shell [8]. These invariant ion settings are Class 1.

Hydronium and hydroxide Born diameters. The H₃O⁺ value 1.2180 Å is directly reported by Figiel et al. for the proton and its $\pm 10\%$ active-bundle response is numerically flat. The OH[−] value 3.0811 Å comes from the retained local hydration-energy inversion and its $\pm 10\%$ response is likewise flat. Neither is an active MEA refinement candidate on the present pressure and speciation evidence, so both are Class 1 for this bundle.

Neutral solvation factors. The active bundle uses $f_{\text{solv},\text{CO}_2} = f_{\text{solv},\text{MEA}} = 1$ and $f_{\text{solv},\text{H}_2\text{O}} = 1.5$. Together with positive ion-specific Born diameters, these values automatically activate corrected SSM+DS. The two unit values make no contribution to the shell average beyond their mole-fraction weight; the water value is the Figiel aqueous-solvent value. Their Class 1 status records the present fixed definition, not universal optimality.

Neutral binary interactions. The current $k_{\text{CO}_2,\text{H}_2\text{O}} = 0.0132628791769$ is the accepted constant fit to 39 qualified Kiepe rows with the induced association fixed [14]. The current $k_{\text{MEA},\text{H}_2\text{O}} = -0.0735274987499$ is the accepted 2B fit to 25 admitted Cai rows [7]. Their calculation records and acceptance limits are owned by ePC-SAFT Issue 80; Class 1 here does not make the full nine-species packet accepted.

Protocol-fixed zero corrections. Binary parameters not separately supplied for the predictive reacting ePC-SAFT

calculation are set to zero in the literature model [19]. Here $k_{ij} = 0$ means that the uncorrected combining rule is used; it does not mean that the physical interaction vanishes. Table 3 therefore assigns Class 1 to literature-zero corrections unless a retained sensitivity result identifies the pair for a specific replay. The bounded CO₂–MEA comparison also retains zero and finds no useful constant correction, so that value is Class 1.

5.2. Class 2: specific comparisons still needed

Non-adopted CO₂ permittivity. The source trail is resolved. Schick et al. define CO₂ as a solvent in their composition mixing rule and report

$$\epsilon_{r,\text{CO}_2}(T) = -0.0036T + 2.467, \quad T \text{ in K},$$

as a correlation developed from three source groups [18, Table 2, p. 4]: Leeke et al., the thermophysical report by Ely et al., and the gas-property measurements by Uhlig et al. The historical scalar is exactly this correlation at 293.0000 K:

$$-0.0036(293.0000) + 2.467 = 1.4122.$$

The earliest retained code copy confirms the same lineage with the adjacent comment “Schick 2022”; the paper’s accepted publication year is 2023.

This resolves the numerical origin but does not validate 1.4122 as a temperature-independent property. The Schick correlation gives 1.4117, 1.3937, 1.3397, 1.2677, 1.1957, and 1.1237 at 293.15, 298.15, 313.15, 333.15, 353.15, and 373.15 K, respectively. Leeke et al. measured CO₂ permittivity at 353 and 383 K over 100–300 bar, while independent measurements show that CO₂ permittivity depends on density and pressure as well as temperature. The Schick equation is therefore an ePC-SAFT modeling correlation, not a universal pure-fluid property.

Decision. The full-domain comparison in Section 11.2 selects solvent-only mass-fraction mixing, which excludes dissolved CO₂ from the dielectric mixing rule. Therefore neither 1.4122 nor the Schick temperature correlation is an active coordinate in Table 1. Both remain retained Class 2 comparison evidence. This decision does not alter the Class 1 molecular Gross–Sadowski row.

Conditional SSM+DS neutral solvation factors. Figiel et al. define f_{mix} using ions and solvents only, set pure-ion factors to one, and fit solvent-specific factors of 1.5, 1.4, and 1.6 for water, methanol, and ethanol, respectively [8, Eqns. 8–10 and Table 2]. They also report that 1.5 represents all three studied solvents at low ion concentration. Because SSM+DS is active, $f_{\text{solv,MEA}}$ is a conditional Class 2 comparison coordinate. It must be profiled jointly with the active ion-specific Born diameters rather than declared independently settled. On 102 common positive pressure observations, increasing it from 1.0 to 1.4, 1.5, and 1.6 reduced pressure log-RMSE from 0.5164 to 0.4243, 0.4028, and 0.3821, respectively, while the 129-target species log-RMSE increased from 0.2333 to 0.2555, 0.2616, and 0.2680. Thus 1.6 gave the best pressure result but 1.0 gave the best species result; no tested value improved both. Because that one-dimensional screen held every $d_{\text{B},i}$ fixed, it ranks only conditional slices through the SSM+DS parameter space; it does not establish a final MEA factor. The bounded automatic-activation screen below confirms that this coordinate is numerically active, but does not identify it independently of the Born diameters.

Figiel et al. explicitly discuss the coupling. In a shell-only model, d_i^{Born} and the shell thickness have the same energetic effect and cannot be considered independently. Their SSM+DS expression adds a dielectric-saturation contribution that depends on d_i^{Born} alone, so the two coordinates become structurally distinguishable, but not automatically well identified from one reactive-MEA response [8, p. 9408]. The authors therefore used a staged fit: ion-specific d_i^{Born} to aqueous infinite-dilution solvation Gibbs energies, solvent-specific f_k to NaBr mean ionic activity coefficients, and ion–water k_{ij} to other aqueous salt activity data. They then iterated all three parameter groups together and reported only minimal changes [8, p. 9411]. This is evidence for a joint profile and final iteration, not for an unconstrained simultaneous fit of both coordinates to MEA pressure alone.

The published mixing rule includes ions and solvents; no other component type was present in Figiel’s systems. The full SSM+DS comparison therefore also removed molecular CO₂ from both the solvation and salt-free dielectric pools. At $f_{\text{solv,MEA}} = 1.6$, exclusion changed common-row pressure log-RMSE only from 0.3821 to 0.3809 and species

log-RMSE from 0.2679669 to 0.2679671, while losing one additional state. This establishes that the CO_2 pool role is immaterial over the tested domain; it does not estimate a standalone CO_2 solvation factor.

Trace-ion molecular rows. The σ_i and ϵ_i/k values for HCO_3^- , CO_3^{2-} , H_3O^+ , and OH^- are the direct Held/Uyan aqueous-ion rows carried into later ePC-SAFT work [12, 19, 20]. They are credible literature values but have not been independently ranked for reactive aqueous MEA.

Water–ion interactions. The literature mapping retains $k_{\text{H}_2\text{O},\text{HCO}_3^-} = 0$, $k_{\text{H}_2\text{O},\text{CO}_3^{2-}} = -0.25$, $k_{\text{H}_2\text{O},\text{H}_3\text{O}^+} = 0.25$, and $k_{\text{H}_2\text{O},\text{OH}^-} = -0.25$ [19, 20]. The bicarbonate zero is a literature assumption, not a confirmed physical exclusion, but no retained comparison identifies it as a present refinement priority, so it follows the Class 1 zero convention. The three nonzero transferred values remain Class 2 pending a controlled current-bundle comparison.

Sensitivity-identified binary comparisons. A 128-sample broad sensitivity survey identified seven zero-valued pairs that should not yet be collapsed into the Class 1 literature convention: $\text{CO}_2\text{--HCO}_3^-$, $\text{CO}_2\text{--CO}_3^{2-}$, MEA--MEA^+ , $\text{H}_2\text{O--MEA}^+$, $\text{MEA}^+\text{--OH}^-$, $\text{MEACOO}^-\text{--H}_3\text{O}^+$, and $\text{CO}_3^{2-}\text{--H}_3\text{O}^+$. They remain Class 2 because the survey used parameter packet 0e7c5ac63c078be9d32c15f316ecdb2b0de67de6f495358ab5955450f0271c6a, not the active bundle. A controlled replay with the active bundle is required before any becomes a Class 3 refinement coordinate. All other zero entries remain Class 1 unless new evidence identifies a material response.

5.3. Class 3: local values and demonstrated refinement candidates

The current $\epsilon_{\text{CO}_2}/k = 173.440\ 25\ \text{K}$ is a 2.5% increase from the Gross–Sadowski start. It was selected jointly with R4 by the bounded campaign in Section 7. Both coordinates are Class 3 because they are project-fit values rather than values copied from a paper.

The May 2026 local-speciation fit varied seven coordinates simultaneously: σ_i , ϵ_i/k , and $d_{\text{B},i}$ for both MEA^+ and MEACOO^- , plus $k_{\text{MEA}^+,\text{MEACOO}^-}$. It used 8 states and 22 residuals from Böttinger, Jakobsen, and Matin data. Overall log-residual RMSE changed only from 0.2714 to 0.2677, and two final evaluations failed. Accordingly, the six displayed ion parameters and the -0.0020 interaction are Class 3 because they are project-fit outputs. Git commit 897bb6e preserves the exact fit values and diagnostics.

Every Debye–Hückel diameter is excluded from the tables and this classification inventory because it is deterministically calculated as $0.88\sigma_i$.

The HCO_3^- and CO_3^{2-} Born diameters of $3.0000\ \text{\AA}$ are also Class 3. The retained 35-state, 51-residual trace-carbonate comparison evaluated paired diameter starts of $1.05\ \text{\AA}$, $3.00\ \text{\AA}$, $4.50\ \text{\AA}$, and $6.50\ \text{\AA}/7.50\ \text{\AA}$. The corresponding overall log-RMSE values were 1.9135, 0.7021, 0.8376, and 1.0614, so the current $3.00\ \text{\AA}$ pair was best among the tested starts. A separate $\pm 0.10\ \text{\AA}$ bicarbonate perturbation produced only small changes. These results provide local numerical support and remove the Class 4 “unscreened default” designation, but the paired coarse scan does not identify either diameter independently. Both remain genuine refinement candidates for a current-bundle factorial comparison against bicarbonate and carbonate observations. Under the active SSM+DS model, that comparison must include a joint $d_{\text{B},i}\text{--}f_{\text{solv,MEA}}$ profile so apparent diameter improvement is not merely compensation by the common solvent-shell factor.

5.4. Class 4: missing values represented by defaults or analogs

Only displayed values with no direct source and no useful sensitivity or comparison evidence belong here. Tables 1–3 currently contain no Class 4 entry. A future placeholder must remain Class 4 until at least a bounded comparison establishes whether it affects the pressure or species results.

The former $c_{\text{MEA}^+} = 2.6000$ and $c_{\text{MEACOO}^-} = 7.0100$ suppression-coefficient defaults are retained only as non-adopted provenance; the active solvent-only rule does not use them and Tables 1–3 do not display them.

The matrix diagonal is gray and unclassified because self-pairs do not use a binary interaction coefficient.

6. Induced-association lineage and correlations

The CO₂ molecular row is not a new induced-association fit. Gross and Sadowski report $m = 2.0729$, $\sigma = 2.7852 \text{ \AA}$, and $\epsilon/k = 169.21 \text{ K}$ in their original PC-SAFT table [9]. Pabsch et al. retain those values, add a 1/1 induced 2B site topology with zero CO₂ self-association energy and $\kappa^{AB} = 0.0450$, and fit the CO₂–water binary term [16, Tables 2 and 6]. Schick et al. carry the same induced row forward [18, Tables 1 and 2]. The retired $m = 2.079$ value is the unsupported value; 2.0729 is the source-correct Gross–Sadowski value.

The retained water correlations are

$$\sigma_{\text{H}_2\text{O}}(T) = 2.7927 + 10.11 \exp(-0.01775T) - 1.417 \exp(-0.01146T), \quad (2)$$

$$\epsilon_{r,\text{H}_2\text{O}}(T) = 78.41043478578715 - 0.36133766233691506(T - 298.15) \quad (3)$$

$$+ 0.00076555618295(T - 298.15)^2. \quad (4)$$

The retained CO₂ scalar is a 293 K snapshot of the Schick correlation $\epsilon_{r,\text{CO}_2}(T) = -0.0036T + 2.467$, not a constant supported across the calculation range [18, Table 2, p. 4]; see Section 5.2.

The current bundle uses the selected **reciprocal induced CO₂–H₂O association set**. CO₂, MEA, and H₂O each carry one donor site *a* and one acceptor site *b*. Upstream PR #132 independently selected MEA 2B over 3B and 4C using the admitted Cai MEA–water rows. The CO₂–water interaction is physical; reactive carbamate formation remains in the reaction network.

Pair	Active edges	$\epsilon^{AB}/k \text{ (K)}$	κ^{AB}	Meaning
MEA–MEA	$a:b, b:a$	2586.30	0.03747	Ordinary MEA self-association.
H ₂ O–H ₂ O	$a:b, b:a$	2425.70	0.04509	Ordinary water self-association.
CO ₂ –H ₂ O	reciprocal a – b cross edges	1212.85	0.04509	Selected induced association: CO ₂ (<i>a</i>)–H ₂ O(<i>b</i>) and CO ₂ (<i>b</i>)–H ₂ O(<i>a</i>).
MEA–H ₂ O	reciprocal a – b	2506.00	0.0411037990944876	Combined cross-association rule.
CO ₂ –CO ₂	none	—	—	Self-association disabled.

7. Corrected replay and parameter-start decision

7.1. Verified packet-construction defect

The upstream reaction audit records the shifts in Eq. (1), but the retained exporter evaluates the raw Austgen R1–R3 correlations and writes those values into records already labeled with the common aqueous-molality standard state. The immutable state packet is preserved unchanged; the notebook replay applies the three shifts before the Engine mapping is constructed. The affected file and Engine commit are identified in Section 12. This distinguishes a packet-export error from a reaction-model or Engine-transfer error.

7.2. Prior ion-specific baseline and reaction screen

The corrected former ion-specific foundation evaluates every one of the 44 speciation states and 76 of the 79 pressure states. Material balances, reaction affinities, and pressure residuals remain below 3.0×10^{-10} in their reported infinity norms. The model therefore solves the stated equilibrium problem accurately where it evaluates; the comparison with observations tests model validity, not nonlinear-solver convergence.

Candidate	N_p	median $p_{\text{model}}/p_{\text{data}}$	pressure RMS factor	N_x	species MAE	species RMS factor
Current R4/R5, corrected R1–R3	76/79	13.89	16.71	44/44	0.00257	1.94
Böttinger R4, corrected R1–R3	70/79	10.25	13.27	44/44	—	2.96

The former ion-specific foundation reproduces the measured species pattern within a median multiplicative factor of 1.065, but its pressure is high by a median factor of 13.89. Böttinger’s common-molality R4 lowers pressure by only 15% on the 70 common evaluable rows, worsens species agreement, and adds six pressure non-evaluations. Its R5 is numerically indistinguishable from Bates–Pinching over the tested range (less than 1.8×10^{-5} difference in $\ln K$). Neither reaction substitution resolves the pressure discrepancy.

The pressure bias is also temperature structured:

T (K)	313.15	333.15	353.15	373.15	393.15
median $p_{\text{model}}/p_{\text{data}}$	26.75	23.71	7.27	11.03	6.82

7.3. Bounded physical-interaction screens

The constant binary screens used 18 pressure states common to every candidate and six representative speciation states spanning the available temperatures.

Coordinate and value	median pressure factor versus current	species scaled RMSE	Interpretation
$k_{\text{CO}_2, \text{MEA}} = -0.1957268589$	0.9978	0.2536	Retain only as Issue #119’s second optimizer start.
$k_{\text{CO}_2, \text{MEA}} = +0.1015997042$	1.0010	0.2542	Old fit used the unconverted reaction packet; not transferable.
$k_{\text{CO}_2, \text{MEA}} = -1$	0.9841	0.2554	Diagnostic bound remains too weak and adds a species non-evaluation.
$k_{\text{CO}_2, \text{H}_2\text{O}} = 0$	0.7838	unchanged	Still leaves a 7.76 median pressure ratio and conflicts with the independent CO_2 –water qualification.
$k_{\text{CO}_2, \text{H}_2\text{O}} = 0.016724$	1.0661	unchanged	Pabsch’s 313 K value worsens reactive pressure relative to the current constant fit.

The verified conclusion from this earlier bounded screen was that no retained or directly transferable literature bundle presently gives both a reasonable nine-species speciation replay and a reasonable CO_2 pressure replay under the selected explicit-ion structure. That screen therefore defined the then-defensible calibration start as the corrected R1–R3 basis, Tong–Aroua R4, Bates–Pinching R5, and $k_{\text{CO}_2, \text{MEA}} = 0$; the legacy negative value was only a second start. Section 11.2 superseded its dielectric choice, and Section 7.4 supersedes its R4 and CO_2 dispersion coordinates.

Baygi’s reported pressure performance cannot be transferred by copying its parameters. That calculation used 3B MEA, 4C water, nonassociating CO_2 , ideal Smith–Missen chemistry, and $k_{\text{MEA}, \text{H}_2\text{O}} = -0.052$ [4]. The present foundation uses 2B MEA, Held 2B water, induced CO_2 –water association, explicit electrolyte activities, and $k_{\text{MEA}, \text{H}_2\text{O}} = -0.0735275$. The two parameter sets do not represent the same model.

7.4. Bounded joint R4– CO_2 campaign and selection

The September 2026 campaign used direct coupled Engine reactive bubble-point and homogeneous-liquid equilibrium solves throughout. A native-regression smoke test on six states and two coordinates required 129 seconds for only two iterations and did not produce a usable fit, so the retained study used parallel, bounded direct evaluations rather than an ad hoc SciPy equilibrium surrogate. The sequence comprised a centered sensitivity screen, R4 pivot–slope grids, complete-data validation, a local R4– CO_2 refinement, and a final slope-boundary check. No numerical score was treated as an accept/reject gate; candidates were compared on error, convergence coverage, temperature residual pattern, and speciation tradeoffs.

The centered screen found no numerical response to the inactive $f_{\text{solv}, \text{HCO}_3^-}$ coordinate, negligible benefit from the tested bicarbonate Born movement, and worse pressure/speciation tradeoffs from the tested R5 direction. R4 was the only consistently useful reaction block. Reparameterizing R4 around 333.15 K separated its local pivot from its temperature slope; a small CO_2 dispersion-energy increase then reduced the remaining common pressure bias.

The selected current bundle is

$$\ln K_{R4} = 3.3515778178 - 1895.3/T, \quad \epsilon_{\text{CO}_2}/k = 173.440 \text{ 25 K.}$$

All other coordinates remain equal to the previous bundle. The final comparison is:

Bundle	pressure coverage	pressure log RMSE	balanced pressure RMSE	speciation coverage / log RMSE
Previous	125/161	0.3974	0.3569	131/131 / 0.2664
Selected	130/161	0.3540	0.3185	131/131 / 0.2658

Pressure log RMSE improves at every temperature: 0.5047 to 0.4445 at 40 °C, 0.2888 to 0.2683 at 60 °C, 0.2491 to 0.2314 at 80 °C, 0.2315 to 0.1971 at 100 °C, and 0.4280 to 0.3809 at 120 °C. The sign pattern remains diagnostic: mean log residuals are +0.4014 and +0.1698 at 40 and 60 °C, then −0.1052, −0.0552, and −0.3661 at 80, 100, and 120 °C. Thus that pre-Born-selection model overpredicted low-temperature pressure and underpredicted high-temperature pressure. Extending the R4 slope to its present bound did not improve the screen and began losing speciation evaluations, so the bounded direction is closed rather than extrapolated farther.

A separate SciPy reaction-extent spot check at `v1e_obs_0128`, using Engine composition derivatives at the Engine bubble pressure, returned to the Engine liquid composition within 3.31×10^{-13} in mole fraction and a maximum reaction-affinity residual of 3.31×10^{-8} . This verifies that spot-check root only; no SciPy-derived parameter score is retained.

8. Nine-species equilibrium replay

Figure 1 is the current smooth-grid visual generated with the exact selected Tables 1–4 bundle and corrected replay. The single 20 °C, 30 wt% MEA panel is the densest retained same-condition comparison: 149 positive values for the five principal species and the MEA+MEA H^+ observation operator from 39 Böttinger, Jakobsen, and Matin states. These include direct, aggregate, inferred, and correlated values. The 17 exact reported-zero records remain in the retained observation snapshot but are omitted from the logarithmic plot. The broken logarithmic scale gives most of the panel to the principal-species range from 5.6248×10^{-5} , 30% below the lowest positive observed or modeled HCO_3^- value, to 0.1134, only 3% above the displayed maximum. A compact 10^{-10} to that cutoff trace panel retains the low CO_2 curve. No retained positive CO_2 or HCO_3^- observation is below 10^{-10} ; the values shown there are calculated molecular CO_2 . CO_3^{2-} , H_3O^+ , and OH^- remain in every equilibrium calculation; the eight reported carbonate values also remain in the figure source snapshot, but trace species are omitted from the visual. Dashed curves connect direct pinned-Engine evaluations from loading 0.075 to 1.0; they are not fitted or interpolated curves.

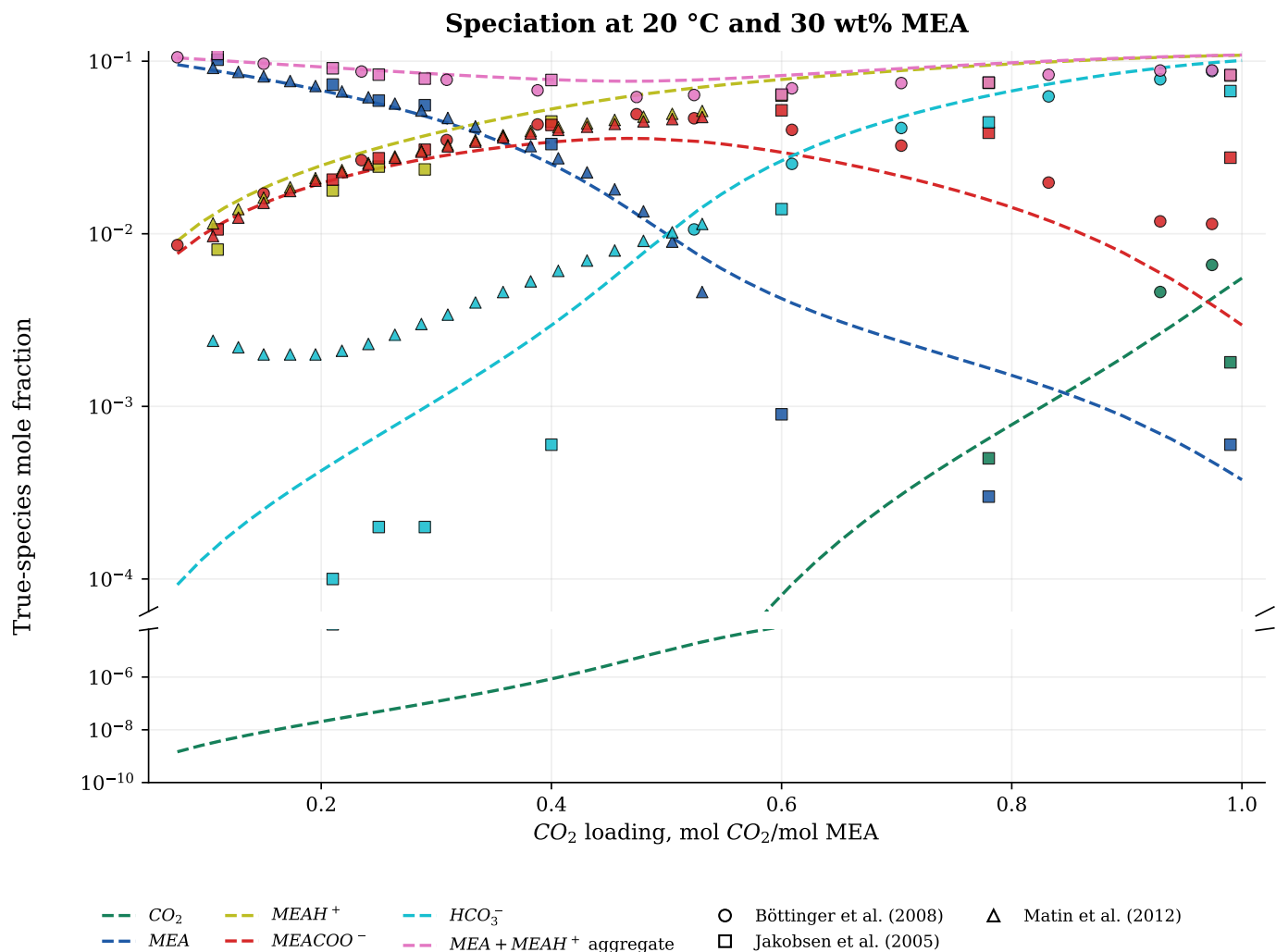


Figure 1. Current smooth-grid diagnostic for the full nine-species liquid-equilibrium replay at 20 °C and 30 wt% MEA. Symbols show all positive retained Böttinger, Jakobsen, and Matin values for the displayed series; symbol shape identifies source and color identifies species. The broken logarithmic axis expands the 5.6248×10^{-5} – 0.1134 principal-species region and compresses the 10^{-10} – 5.6248×10^{-5} range. Dashed lines connect direct Engine evaluations of the selected bundle through loading 1.0.

These states are homogeneous reactive-liquid solves, not ideal-reaction or postprocessed SciPy curves. Each state couples the five reaction constraints to ePC-SAFT chemical potentials for all nine species at fixed temperature and pressure. A 2026-09-01 disposable timing check on this machine, excluding process startup and model construction, measured 0.935 s/state for a warm five-state second pass with the retained wheel. The same five feeds took 2.260 s/state with a disposable wheel built from Engine main after only the required provider-to-EOS request-schema renames. This is a route-specific diagnostic, not a general Engine benchmark: recent GREPE and regression speedups do not establish a speedup for this homogeneous-liquid owner.

9. Carbon-dioxide pressure replay

Figure 2 shows all 161 active-v1 30 wt% MEA pressure observations from Aronu, Hilliard, Idris, Jou, Mamun, and Xu at 40–120 °C. The selected replay evaluates all 161 states. Its full-data $\log_{10}$ RMSE is 0.3317. Lines interpolate the 161 evaluated selected-bundle observation states; no additional smooth-grid Engine campaign was required, and every experimental point remains shown.

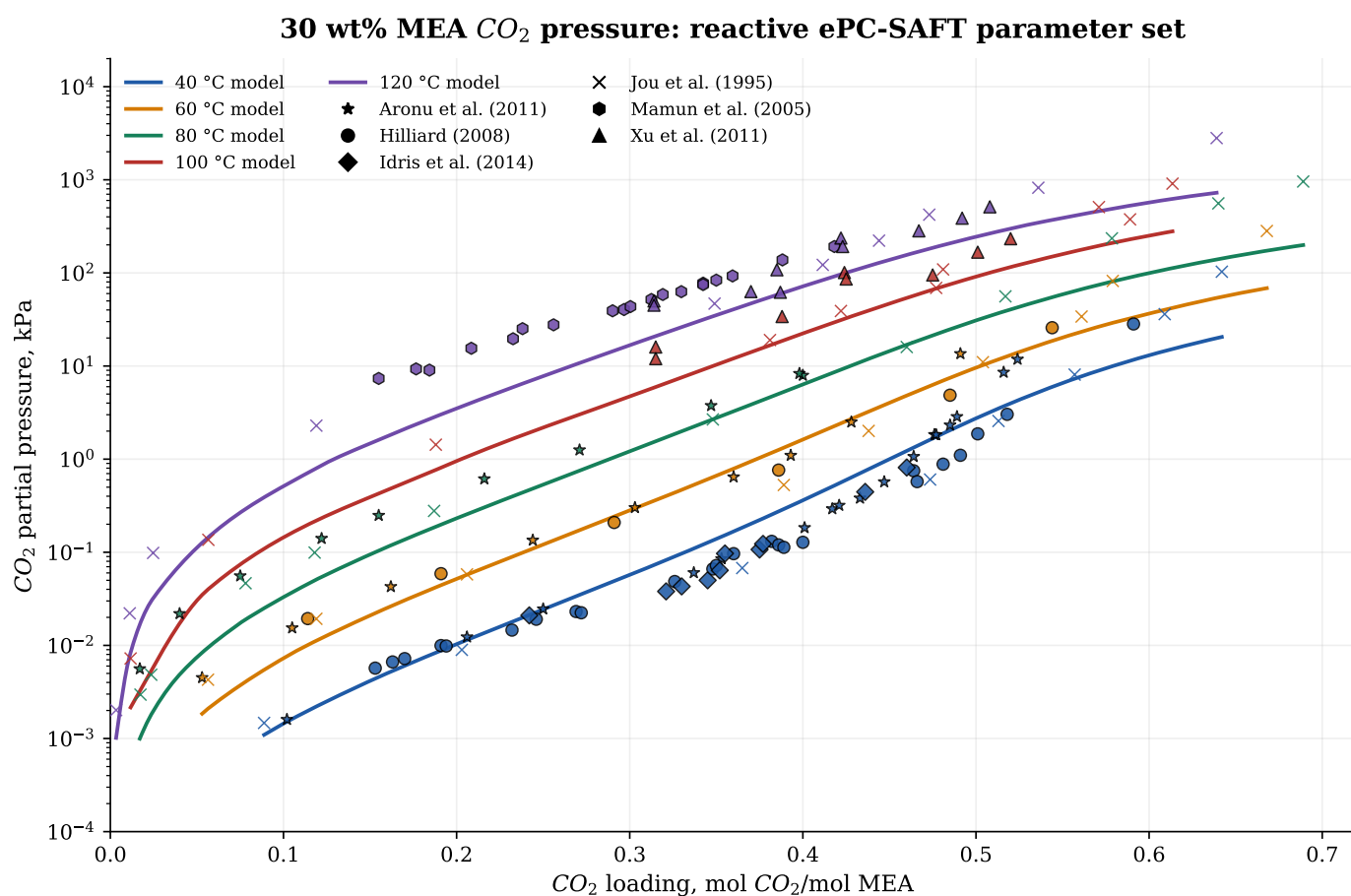


Figure 2. Carbon-dioxide partial pressure for 30 wt% MEA. Colors identify temperature, symbols identify experimental source, and lines are shape-preserving log-response interpolations of newly calculated active-bundle states within each evaluated loading range; every retained observation and corresponding model state is included.

The 41 pressure non-evaluations reported by the earlier independent replay were branch-selection failures, not missing thermodynamic solutions. That replay rebuilt every row from its source start and discarded the nearest certified liquid state. In a representative 120 °C failure, the inner reactive liquid passed its balance, reaction, pressure, KKT, and local stability checks, while the outer bubble iteration alternated between about 272 and 739 kPa until its iteration limit. Projecting a certified neighboring loading or temperature state onto the target balances places the same fixed parameter problem on the continuous liquid branch. The deterministic replay now tries that nearest certified branch, retains the

original independent start as a fallback, and uses a neighboring-temperature certified branch only when needed. It recovers all 41 rows without changing parameters, tolerances, or iteration limits.

9.1. Current-bundle fit and coverage statistics

These are replay/agreement statistics for the exact bundle in Tables 1–4. Log residuals are $\log_{10}(y_{\text{model}}/y_{\text{obs}})$; the RMSE factor is $10^{\text{RMSE}_{\log_{10}}}$. Native pressure errors are in kPa and native speciation errors are mole fractions.

Family	Coverage	log RMSE	RMSE factor	MAPE (%)	MAE	Native RMSE
Pressure	161/161	0.3317	2.146	55.75	48.1848	195.5063
Speciation targets	125/131	0.3702	2.345	23.26	0.005180	0.008568

Pressure performance and, especially, evaluability vary strongly by temperature:

T (°C)	Coverage	log RMSE	RMSE factor	MAPE (%)	Median APE (%)
40	59/59	0.2698	1.861	66.71	63.85
60	24/24	0.2507	1.781	41.02	38.67
80	21/21	0.4069	2.552	49.55	58.10
100	18/18	0.3422	2.199	46.56	48.38
120	39/39	0.4027	2.527	55.83	58.43

The principal measured-species breakdown shows that bicarbonate, not the amine/carbamate coordinates, dominates the speciation log score:

Target	Coverage	log RMSE	RMSE factor	MAPE (%)
MEA	18/19	0.0582	1.143	9.70
MEA H^+	18/19	0.0943	1.243	23.17
MEA + MEA H^+	24/25	0.0598	1.148	9.11
MEACOO $^-$	42/44	0.4901	3.091	14.65
HCO $_3^-$	23/24	0.5412	3.477	64.42

The complete machine-readable handoff is retained in the results directory:

- row-level observed/predicted values: `current-best-fit-residuals.csv`;
- grouped overall, temperature, source, species, and temperature–source statistics: `current-best-fit-statistics.csv`;
- failed coordinates and diagnostics: `figure-calculation-failures.csv`.

The bundle can be used for absorption-column calculations. All 161 retained pressure observations evaluate, including all 39 at 120 °C. Two observed speciation states and three optional smooth-grid states remain non-evaluable and are explicit in the failure table.

10. What pressure constrains and what remains unresolved

At phase equilibrium the molecular carbon-dioxide fugacities satisfy

$$f_{\text{CO}_2}^V = f_{\text{CO}_2}^L. \quad (5)$$

An activity/Henry representation writes the same comparison schematically as

$$y_{\text{CO}_2} \phi_{\text{CO}_2}^V P = x_{\text{CO}_2} \gamma_{\text{CO}_2}^L H_{\text{CO}_2}. \quad (6)$$

inference: pressure observations therefore constrain the liquid molecular- CO_2 chemical potential together with loading, material balances, and the reaction network. They are valuable indirect speciation evidence, but they do not separately identify free x_{CO_2} , liquid nonideality, reaction constants, association, binary interactions, dielectric terms, or solvation factors. Direct species observations and pressure observations are combined because they constrain different projections of the same equilibrium state; pressure is not a substitute for direct speciation.

verified: the retained replay does not assume an ideal vapor. Its state packet declares a neutral incipient-vapor phase and reads `phase.partial_pressure` from the pinned ePC-SAFT equilibrium result. By contrast, the current public IDAES column property files declare an ideal liquid/Henry-law MEA solvent and an ideal vapor; they are not the modified-eNRTL model from a separate literature study. The public solvent file retains the three apparent neutral components CO_2 , MEA, and H_2O plus the three ionic species MEA^+ , MEACOO^- , and HCO_3^- [official IDAES solvent file]; the vapor file explicitly selects the ideal equation of state [official IDAES vapor file].

10.1. Reduced-chemistry diagnostic

The six-species reduction retains CO_2 , MEA, H_2O , MEA^+ , MEACOO^- , and HCO_3^- . To isolate species-set structure without importing IDAES empirical equilibrium constants, derive its two net constants from the current five-reaction basis:

$$K_{\text{carbamate}} = \frac{K_2}{K_4 K_5}, \quad \text{CO}_2 + 2 \text{MEA} \rightleftharpoons \text{MEACOO}^- + \text{MEA}^+, \quad (7)$$

$$K_{\text{bicarbonate}} = \frac{K_2}{K_5}, \quad \text{CO}_2 + \text{MEA} + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{MEA}^+. \quad (8)$$

The first relation is the exact R2–R4–R5 projection already recorded in MEA Issue #72. This makes the comparison a reaction-space projection of the same model rather than a change of both chemistry and equilibrium constants.

verified: across the 44 retained nine-species calculations, the largest mole fractions are 1.0307×10^{-2} for HCO_3^- , 1.1687×10^{-4} for CO_3^{2-} , 3.3641×10^{-6} for OH^- , and 8.9480×10^{-12} for H_3O^+ . The largest omitted net-charge contribution $(2x_{\text{CO}_3^{2-}} + x_{\text{OH}^-} - x_{\text{H}_3\text{O}^+})/x_{\text{MEA}^+}$ is 0.01949 at Böttinger state 030. These values are calculated directly from `figures/speciation/output/speciation-model.csv`.

The complete matched comparison is retained under `../enrtl_six_species_ideal_comparison/`; its Quarto source and rendered PDF are the detailed calculation record. In its ideal-liquid lane, all 232 unique canonical coordinates and 464 model rows converged with a maximum equation residual of 1.9895×10^{-13} . The largest difference among the six shared species was 6.5100×10^{-3} mol per mol initial MEA, and carbonate dominated the omitted inventory. In its fixed- T , P Engine/SciPy lane at 100 kPa, the carbonate amounts at 293.15, 313.15, and 323.15 K were 2.99×10^{-4} , 1.56×10^{-4} , and 1.10×10^{-4} mol per mol initial MEA; the largest shared-species changes were approximately 2.9×10^{-4} mol per mol initial MEA. That lane is a liquid-composition calculation, not a bubble-pressure calculation.

verified structural result: when the six- and nine-species states are both projected to the same conserved apparent totals, their neutral apparent pressure inputs agree to 1.321165×10^{-14} . The neutral `flashTQ` route converged for only 1 of 161 pressure observations and predicted 2038.38 kPa for an observed 0.002 02 kPa. A neutral apparent flash therefore cannot reveal the internal reaction or ionic parameters; the pressure calculation must keep the reactive species allocation coupled to the EOS.

The companion sensitivity survey evaluated 143 parameter identities grouped into 113 continuous directions with 128 Latin-hypercube samples, of which 123 were usable. Its strongest recurring directions were R5 a_K , R4 b_K , the water segment-diameter temperature terms, MEA^+ dispersion and ionic terms, and selected CO_2 –water, carbonate–water, MEA– MEA^+ , MEA^+ –water, carbamate–hydronium, and bicarbonate– CO_2 binary interactions. Those numerical ranks belong to parameter packet 0e7c5ac63c078be9d32c15f316ecdb2b0de67de6f495358ab5955450f0271c6a and Engine-wheel hash 05cca73463b9b44dc3a77dfd239b767ec63aafce7d18b85b2a66b80773847a5; the separate fixed- T , P

lane used packet 12e380a27eb615e25bae580f8c1ec852020879917b08fa16a5821ae76311550d. Neither is the active bundle in Tables 1–3, so the ranking must be replayed with the active parameter file before its numerical ordering is used in the manuscript.

inference: the three omitted trace ions have small direct abundance over this replay, which makes a six-species ablation useful as a cheap conditioning test. It does not prove that they are dispensable: the z_i^2 weighting gives carbonate four times the per-mole ionic-strength contribution of a monovalent ion, and the omitted species preserve carbonate, autoprotolysis, pH, charge, and extrapolation structure. Removing them also removes weakly identified ionic parameters, so an apparent fit improvement may come from reduced flexibility rather than better physics.

Retained finding	Present use	Remaining action
Exact R2–R4–R5 and R2–R5 projection	Methods description for a controlled six-species ablation	None for the algebra; cite the retained reaction table.
Ideal and fixed- T , P species comparison	Evidence that carbonate is the dominant omitted species and that the reduction is not simple deletion	Repeat the coupled bubble calculation with the active bundle before using pressure-error conclusions.
Neutral apparent-pressure invariance	Methods/diagnostic explanation of why total absorbed carbon cannot be passed directly to a neutral CO_2 flash	Keep as a route-rejection result; do not present the 1/161 flash as a model fit.
Packet-specific sensitivity survey	Prioritizes the first active-bundle sensitivity directions	Replay the same screen with the active parameter mapping and wheel before reporting ranks or intervals.

11. Interpretation

The R4– CO_2 update reduces pressure error while preserving complete observed- speciation coverage, but the residual changes sign with temperature and all 39 pressure states at 120 °C now evaluate. The remaining temperature-dependent mismatch is therefore a model residual rather than a coverage artifact; it does not justify opening R1–R3, R5, or the ion block as compensating coordinates.

The Class 1 $k_{\text{CO}_2, \text{MEA}} = 0$ uses the uncorrected combining rule; it does not assert an absence of physical interaction. The bounded screen shows that one constant correction is too weak and too temperature uneven to explain the observations, so this coordinate is not a present refinement priority. Likewise, loaded-MEA pressure alone cannot identify $f_{\text{solv}, \text{CO}_2}$ or $f_{\text{solv}, \text{MEA}}$ because those working assumptions are confounded with cross association, dielectric response, reaction, and other physical interactions.

Historical diagnostics remain outside the bundle:

- legacy $k_{\text{CO}_2, \text{MEA}} = -0.1957268589$;
- alternate MEAH^+ $\sigma = 2.753332853$ and MEACOO^- $\sigma = 3.017079292$; and
- trace-only bicarbonate Born-diameter movement toward 6.802 94 Å, with CO_3^{2-} left near 2.997 44 Å.

These values identify sensitivity or lineage; none is adopted. The trace-carbonate result does not establish independent identifiability.

11.1. Source-derived regression rules to preserve

The following rules are retained because they change how a future MEA fit must be designed. They are methodological constraints from the ePC-SAFT lineage, not additional fitted results.

Select the formulation before estimating its parameters. Bülow and Ascani show that the Born expression, composition-dependent permittivity, and their complete composition derivatives change electrolyte activities and phase parti-

tioning; Ascani’s solvent-mass and combined rules can reverse a failed salting-out prediction without introducing ion–solvent corrections [5, Secs. 3.2–4][3, pp. 1310–1313]. Therefore a parameter vector fitted under one Born/permittivity formulation cannot be treated as an unchanged control under another. The fixed-parameter structural screen in Section 11.2 must precede any refit.

Keep parameter blocks with the model choices that generated them. Pabsch et al. explicitly retain the 2B water model because the inherited ion parameters were estimated with that water parameterization; changing the water association scheme would require refitting the ion block [16, p. 5770]. The same restriction applies to the reciprocal induced CO₂–H₂O topology: a k_{ij} obtained with non-induced CO₂ is not interchangeable with the present induced-association value.

Baygi and Pahlavanzadeh additionally show why pure-fluid agreement cannot select an MEA association topology: their 2B, 3B, and 4C MEA fits gave similar pure saturation behavior but different aqueous binary VLE and excess-enthalpy behavior [4, pp. 789, 795–799]. Their nonionic PC-SAFT study selected 3B MEA with 4C water, but that result does not authorize replacing the present 2B/2B reactive ePC-SAFT block. It instead requires any future topology comparison to refit the complete coupled block and judge binary and caloric observables, not pure density and vapor pressure alone.

Assign each coordinate an identifying observable. The literature’s useful hierarchy is: neutral pure-component parameters from density and vapor pressure; neutral binary and cross-association corrections from binary VLE; ionic short-range parameters and ion–water corrections from aqueous density, osmotic coefficients, or MIACs; d_i^{Bom} from ion-specific infinite-dilution solvation evidence; and solvent f_{solv} from finite-concentration MIACs. Figiel then performs one joint iteration and finds only minimal movement [8, p. 9411]. Reactive MEA pressure may test the assembled bundle, but it should not be the sole datum identifying every block.

Keep reaction constants, activities, and standard states together. Uyan et al. solve chemical and phase equilibrium with ePC-SAFT activities and show that replacing species activities by mole fractions worsens the species distribution [19, pp. 96–98]. Therefore R1–R5 coefficients cannot be transplanted, refitted, or compared independently of their concentration scale, reference states, and activity-coefficient convention. Pressure is an indirect reaction constraint; admitted speciation and caloric observations are needed to prevent an EOS parameter from compensating for an inconsistent reaction basis.

Use simpler electrolyte subsystems to resolve reactive-mixture gaps. Uyan et al. set the unsupported water–MDEAH⁺ correction to zero when no pseudo-binary data were available [19, p. 95]. Wangler et al. later found water–HCO₃[−] and water–MDEAH⁺ influential, estimated them from aqueous NaHCO₃ and protonated-MDEA osmotic-coefficient data, and then predicted the reactive CO₂–water–MDEA VLE without fitting that ternary pressure data [20, pp. 19–24]. The MEA analogue is to seek aqueous bicarbonate and protonated-MEA salt activity or osmotic evidence before using reactive pressure to adjust the corresponding binary terms.

Do not call calibration rows predictions. Uyan and Wangler reserve “prediction” for reactive ternary gas-solubility results after parameters were obtained from pure fluids and binary or pseudo-binary subsystems [19, 20]. Any future use of MEA pressure or speciation rows in the objective must label those rows calibration and report performance on held-out temperatures and sources separately.

Compare model error with experimental disagreement. Pabsch et al. screen conflicting CO₂-solubility sources before fitting and compare the model AARD with the inter-source ARSD; similar magnitudes indicate that the experimental record, rather than only the model, limits apparent accuracy [16, pp. 5771–5775]. The MEA campaign should retain source-wise residuals and duplication structure rather than assigning equal confidence to every digitized row.

Treat single-ion solvation values as convention-dependent anchors. Bülow et al. note that individual-ion hydration Gibbs energies are not directly measurable and are assembled through extra-thermodynamic conventions [5, Sec. 3.2]. They are useful for ranking and anchoring Born diameters, but agreement with one tabulation is not independent proof of a unique physical radius. Salt-level MIAC, transfer, density, and osmotic data remain necessary checks.

Bind every result to the exact corrected parameter packet. The Bülow corrigendum reports a parameter error in the

published MIAC calculations and replaces Figs. 6–7 while leaving the qualitative conclusions unchanged [6]. This is direct precedent for retaining source corrections, unrounded parameter files, Engine identity, and hashes; a figure or rounded table alone is not a reproducible parameter authority.

Use caloric behavior as a reaction-model check. Wangler et al. use absorption enthalpy as a downstream prediction and connect its quality to the reaction equilibria and the temperature and composition dependence of species activities [20, pp. 24–26]. After pressure and speciation are stable, retained MEA absorption-enthalpy data provide an independent test of R1–R5 that pressure-only refinement cannot supply.

11.1.1. Absorption-enthalpy observations and required calculation

The repository already contains the observations needed for a paired-state calorimetry study; no data import from the eNRTL fitting repository is needed. The canonical table `MEA_heat_of_absorption_observations.csv` is retained under `data/reference/MEA/observations/calorimetry/`. Its exact SHA-256 is recorded in the calculation receipt. The materialized campaign view preserves every source field and adds the partition, pair group, preceding endpoint, and two required sensitivity labels.

Source and temperature	Rows	Role	Use
Kim–Svendsen 2007, 40 and 80 °C	66	Calibration	Available for a bounded joint pressure–species–enthalpy fit.
Kim–Svendsen 2007, 120 °C	20	Temperature holdout	Evaluate after model selection; do not use it to pull the temperature fit.
Kim et al. 2014, Table A1-1	27	Model-selection comparison	The earlier eNRTL study inspected these rows, so they are not untouched validation for that candidate.

The Engine now supplies complete reactive total enthalpy on the solved-pressure bubble-point route. The current bundle was therefore replayed without fitting against all 113 retained calorimetry observations. All 113 endpoint states and all 113 finite loading intervals evaluated. The overall RMSE is 23.98, the mean bias is −18.57, and the median absolute error is 19.25 kJ mol^{−1} CO₂. The calibration partition gives RMSE 20.01, the Kim et al. model-selection comparison gives 16.84, and the 120 °C temperature holdout gives 39.30 kJ mol^{−1} CO₂. The fixed bundle therefore systematically underpredicts heat release, with the largest disagreement at the held-out high temperature.

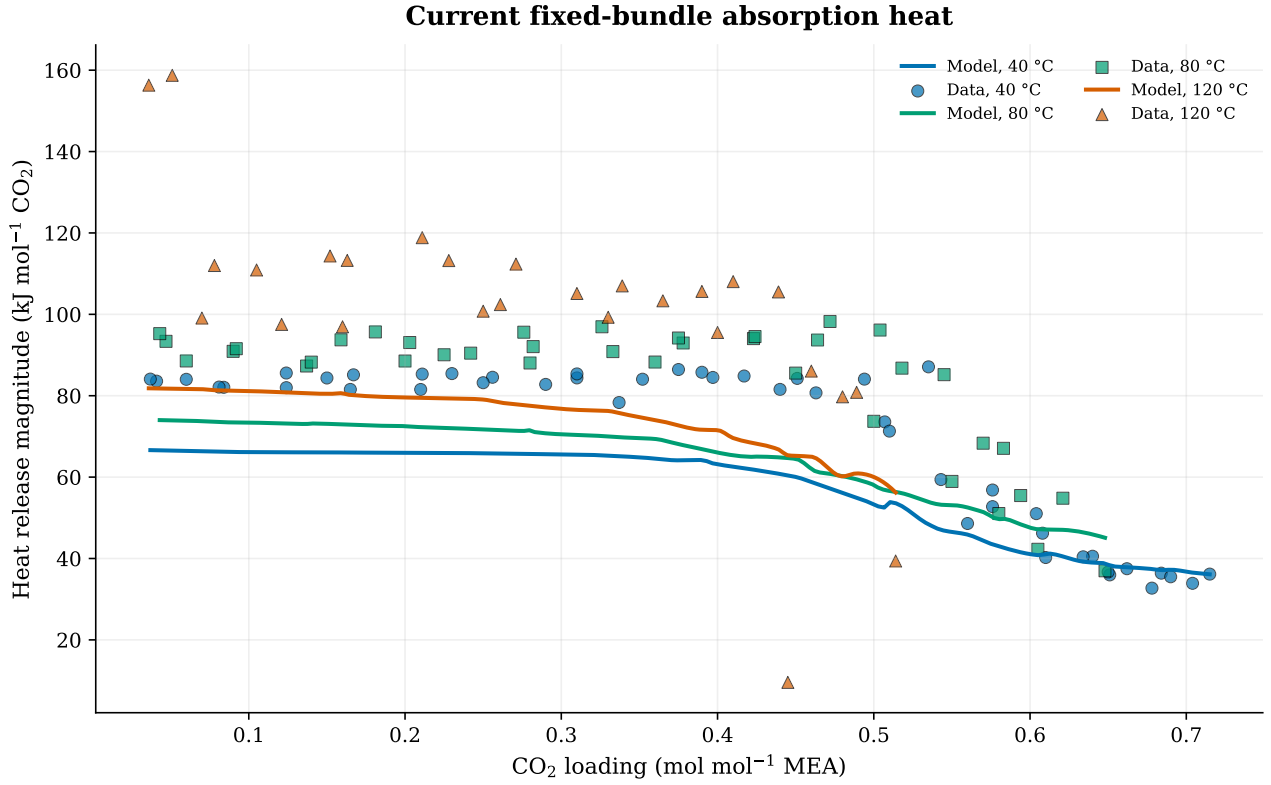


Figure 1: Retained calorimetry observations (points) and the current fixed Tables 1–4 bundle evaluated by direct total reactive enthalpy (continuous lines). Lines are shape-preserving interpolations of the scored finite-interval predictions; no calorimetry parameter was fitted.

For adjacent loading endpoints $i - 1$ and i in one source, temperature, MEA concentration, and experimental run, the modeled positive heat-release magnitude must be

$$q_i^{\text{model}} = -\frac{n_i h_{L,i} - n_{i-1} h_{L,i-1} - \Delta n_{\text{CO}_2,i} h_{\text{CO}_2,\text{ig}}(T)}{\Delta n_{\text{CO}_2,i}}. \quad (9)$$

Here nh_L is total equilibrium-liquid enthalpy, not residual enthalpy alone; $h_{\text{CO}_2,\text{ig}}$ is incoming ideal-gas carbon-dioxide enthalpy on the same reference basis; and the leading minus sign converts an exothermic enthalpy change to the reported positive release magnitude in $\text{kJ mol}^{-1} \text{CO}_2$. The first modeled endpoint in each series uses the declared starting-loading assumption $\alpha_{\text{CO}_2} = 0.003$; that assumption and the likely source typo at kim2007_t40_r1_0.041 require separate exclusion or perturbation checks [15, Table 1, p. 3].

This is a direct finite-dose energy balance, not a pressure-slope estimate. For each endpoint the same nine-species reactive liquid is solved, and the Engine combines declared species reference enthalpies with the active ePC-SAFT residual departure. Reaction heat is not added as a separate empirical correction: it enters through the equilibrium species amounts on one common standard-state enthalpy basis.

The caller-owned reference thermochemistry is constructed without calorimetry fitting. At 313.15, 353.15, and 393.15 K, the Engine evaluates the source-to-EOS temperature transfer and the typed R1–R5 correlations give

$$\Delta_r h_r^\circ = RT^2 \left(\frac{\partial \ln K_{r,\text{EOS}}}{\partial T} \right), \quad \mathbf{h}^\circ = \mathbf{S}^\top (\mathbf{S} \mathbf{S}^\top)^{-1} \Delta_r h^\circ. \quad (10)$$

The second relation chooses a deterministic minimum-norm conserved gauge. The 393.15 K calorimetry results extrapolate R5 beyond its stated 323.15 K correlation domain, so the large 120 °C disagreement is evidence for the next reaction-temperature study rather than a validated high-temperature R5 prediction. Quadratic interpolation through the three temperature knots supplies one reference enthalpy and linear heat-capacity polynomial per species. The same

thermochemistry fingerprint is used at every loading, and the Engine independently checks $\text{Sh}^\circ = RT^2 \partial \ln \mathbf{K} / \partial T$ before returning an enthalpy.

The retained wheel is built from Engine commit 8438ce5f94a547189c91c4ec180a7782d60879d6. Upstream PRs 199–201 add reference-consistent total enthalpy, correct the reference-pressure derivative, and preserve reaction-temperature derivatives through solved-pressure and incipient-boundary reconstruction. The final replay required 191 solver attempts: 78 initial starts failed before a certified neighboring-loading or neighboring-temperature continuation recovered the same state. No tolerance, iteration limit, parameter, or equation was changed, and no endpoint remains failed. The earlier high-temperature gaps were initialization-path failures in the outer bubble-pressure route, not absent thermodynamic solutions.

The present reconstruction assumes zero finite vapor inventory in the calorimeter and incoming ideal-gas CO_2 with zero residual enthalpy on the same gauge. Those assumptions are explicit in the retained receipt. The older Gibbs–Helmholtz diagnostic remains useful as an independent pressure-slope comparison, but it is no longer the primary calorimetry result. Both methods show the same negative heat-release bias. The direct result identifies a real model deficiency, but it does not by itself distinguish reaction temperature coefficients from missing reference heat-capacity or residual-enthalpy structure. The next fit should therefore use the smallest supported R1–R5 enthalpy-coordinate subset and judge it simultaneously against pressure, speciation, and the untouched 120 °C holdout.

The calorimetry view is retained at `data/input/calorimetry-observation-partition.csv` within this analysis. Its receipt records 113 rows in nine pair groups, positive loading increments, the canonical and selected-view hashes, and the two sensitivity requirements. This prepares the evidence without changing the active parameter mapping.

11.1.2. Current bundle against those rules

The current bundle is usable as the best retained working bundle, but it is not equally mature in every parameter block. “Meets” below means that the present lineage and experiment are appropriate; it does not mean that the numerical value can never be revised.

Block	Assessment	Present evidence	Consequence
Model structure	Meets	2B water and MEA, reciprocal induced $\text{CO}_2\text{-H}_2\text{O}$ association, solvent-only mass-fraction permittivity, and automatic SSM+DS were evaluated as one declared structure.	Keep automatic activation fixed while profiling the active Born diameters and neutral solvation factors together.
Neutral molecular block	Mostly meets	Water and MEA coordinates retain literature lineages; $k_{\text{MEA,H}_2\text{O}}$ and $k_{\text{CO}_2,\text{H}_2\text{O}}$ were fitted to their binary systems with the active association choices.	Do not reopen the neutral block broadly. The locally adjusted ϵ_{CO_2}/k must be called a reactive-MEA calibration coordinate and checked on held-out sources.
Reaction block	Partial	R1–R5 use a coherent standard-state conversion, but R4 was adjusted with ϵ_{CO_2}/k against reactive pressure and the older sensitivity survey identified R5 as influential.	Recompute active-bundle R4/R5 sensitivities and use pressure, species, and direct absorption enthalpy together; do not infer either reaction solely from pressure.
Principal amine ions	Needs stronger anchors	The MEA^+ and MEACOO^- molecular and Born coordinates are local Class 3 fit outputs from 8 states and 22 residuals, with only a small total improvement and two failed final evaluations.	Seek protonated-MEA salt and carbamate-relevant electrolyte evidence before another broad reactive-mixture fit.
Carbonate and trace ions	Partial	Literature molecular rows are retained, but the $\text{HCO}_3^-/\text{CO}_3^{2-}$ Born diameters have only a paired coarse screen and the OH^- diameter is a local inversion. Bicarbonate log-RMSE is 0.6154, far larger than the other measured species.	Prioritize bicarbonate chemistry, the carbonate Born radii, and their water interactions in a controlled active-bundle identifiability screen.
Binary interaction matrix	Mixed	Literature-zero corrections are fixed, three nonzero water–trace-ion transfers remain Class 2, and seven zero pairs were sensitive only in an older packet.	Replay only those named Class 2 directions with the active packet before promoting any k_{ij} to a fit coordinate.

Evidence lane	Assessment	Present evidence	Consequence
Pressure and numerical coverage	Complete for observations	Pressure log-RMSE is 0.3317 for 161/161 states, including 39/39 at 120 °C. The mean residual is positive at 40 °C and negative from 60 through 120 °C.	Use all observations in later model comparisons; do not reinterpret the recovered rows as new experimental evidence.
Speciation	Mixed	125/131 positive observed targets evaluate. MEA and MEAH ⁺ remain comparatively strong at log-RMSE 0.0582 and 0.0943, while MEACOO ⁻ and HCO ₃ ⁻ are weak at 0.4901 and 0.5412.	Preserve the free/protonated-amine behavior and target carbamate–bicarbonate coupling rather than globally refitting every ion.
Permittivity and solvation	Conditional	The active bundle uses solvent-only mass-fraction mixing with automatically activated SSM+DS, but no independent loaded-MEA static-permittivity data validate the predicted response.	Obtain that evidence and jointly profile $f_{\text{solv,MEA}}$ with the uncertain Born diameters.
Calibration separation	Partial	Source-wise residuals, hashes, and exact packets are retained, but the selected R4–CO ₂ update used the current reactive observations.	Freeze explicit calibration and held-out source or temperature partitions before the next parameter optimization.
Absorption enthalpy	Complete fixed-bundle replay	The Engine returns reference-consistent total reactive enthalpy, and all 113 retained intervals evaluate. Overall RMSE is 23.98 and the 120 °C holdout RMSE is 39.30 kJ mol ⁻¹ CO ₂ .	Use the systematic negative bias to constrain a small reaction-temperature coordinate set while preserving the current pressure and speciation performance.

Current diagnosis. The neutral and association foundation is the strongest part of the bundle, and free/protonated-amine speciation is described well. The immediate scientific weaknesses are bicarbonate and carbamate, followed by insufficiently anchored ionic/Born coordinates. The former high-temperature bubble-pressure coverage weakness has been removed for the retained pressure observations by preserving certified loading and temperature continuation states. Three optional smooth speciation-grid points remain non-evaluable. Solver coverage and scientific fit must remain separate: a recovered solve is not evidence for a different parameter value, and a converged residual is not evidence that the ionic block is identified.

11.2. Full-domain permittivity selection

This fixed-parameter structure study asks how the current solvent-only mass-fraction rule with automatically selected SSM+DS compares with the previous original-Born baseline and other source-consistent structures in its combined description of reactive MEA pressure, speciation, dielectric response, and physical branch identity. It changes no reaction, molecular, ionic, association, binary, vapor, tolerance, or initialization input. Reciprocal induced CO₂–H₂O association and the unique Born diameters in Table 2 remain active in every calculation. No parameter was fitted.

11.2.1. Equations, pools, units, and derivatives

Let $N = \{\text{H}_2\text{O}, \text{MEA}, \text{CO}_2\}$, I be the six charged species in Table 2, $x_N = \sum_{j \in N} x_j$, and $w_j^N = x_j M_j / \sum_{k \in N} x_k M_k$. Rule A intentionally uses the explicit solvent pool $N_A = \{\text{H}_2\text{O}, \text{MEA}\}$. Rules B–E include molecular CO₂ explicitly. No rule

silently removes or renormalizes another component. The screened bulk relative-permittivity rules are

$$\epsilon_r^A = \sum_{j \in N_A} w_j^{N_A} \epsilon_{r,j}, \quad \text{solvent-only mass-fraction mixing,} \quad (11)$$

$$\epsilon_r^B = \sum_{j \in N \cup I} x_j \epsilon_{r,j}, \quad \text{all-component mole-fraction mixing,} \quad (12)$$

$$\epsilon_r^C = \frac{\sum_{j \in N \cup I} x_j M_j \epsilon_{r,j}}{\sum_{j \in N \cup I} x_j M_j}, \quad \text{all-component mass-fraction mixing,} \quad (13)$$

$$\epsilon_r^D = x_N \sum_{j \in N} w_j^N \epsilon_{r,j} + \sum_{i \in I} x_i \epsilon_{r,i}, \quad \text{combined solvent-mass/ion-mole mixing,} \quad (14)$$

$$\epsilon_r^E = \frac{\epsilon_{r,N}^{\text{salt-free}}}{1 + 7.01 x_I}, \quad x_I = \sum_{i \in I} x_i, \quad \text{nonlinear ionic suppression.} \quad (15)$$

Rules B–D use $\epsilon_{r,i} = 8$ for every ion. The pure-component inputs are the retained water correlation, $\epsilon_{r,\text{MEA}} = 32$, and the Schick transfer $\epsilon_{r,\text{CO}_2}(T) = 1.4122 - 0.0036(T - 293 \text{ K})$ [18, Table 2]. Equations 11–14 use the three linear mixing forms and salt-free-solvent definition reported by Ascani and Held [3, Eqns. 11–13]; Rule A retains only the normalized salt-free-solvent average within Equation 13. Equation 15 is the nonlinear empirical electrolyte-suppression rule fitted at 298.15 K [8, Eqns. 11–12]. Relative permittivity, f_{solv} , c_{shell} , and $c_{\text{dielectric}}$ are dimensionless; d_i^{Born} is stored in Å and converted to metres. The exact source locators are Bülow’s original-Born Eq. 2 and composition-derivative Eqs. 3–5, with the linear dielectric discussion around Eq. 7; the Bülow corrigendum’s correction to the input used in Figs. 6–7; Ascani’s Born Eq. 10 and mixing Eqs. 11–13; and Figiel’s SSM+DS Eqs. 6–10, nonlinear dielectric Eqs. 11–12, and Supporting Information Table 1, Table S2, and Figs. S1–S6. The corrigendum reports that the corrected figures do not change Bülow’s conclusions. The full receipt records separate SHA-256 identities for the Bülow article, its corrigendum, the Ascani article, the Figiel article, and the Figiel Supporting Information rather than treating any pair as one source.

Rule A applies the normalized salt-free-solvent average to water and MEA. Dissolved molecular CO_2 and all ions therefore have zero direct weight in this rule. This descriptive name identifies the equation and component pool without attaching a system-specific parameter-set name to it.

For rules with temperature-dependent pure-component values, the exact Engine automatic derivatives follow the declared algebra. For example, $\partial \epsilon_r^B / \partial x_i = \epsilon_{r,i}$ and, if $\epsilon_r^C = N_C / M_C$, $\partial \epsilon_r^C / \partial x_i = M_i(\epsilon_{r,i} M_C - N_C) / M_C^2$. The corresponding temperature derivatives sum the pure-component $\partial \epsilon_{r,j} / \partial T$ terms; all five rules have $\partial \epsilon_r / \partial \rho = 0$. Rules D and E use exact quotient and chain derivatives rather than frozen-composition approximations.

The original Born residual is

$$\frac{a^{\text{Born}}}{RT} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \sum_{i \in I} x_i z_i^2 \frac{1 - 1/\epsilon_r}{d_i^{\text{Born}}}. \quad (16)$$

The corrected Engine SSM+DS form first defines

$$d_i^{\text{out}} = d_i^{\text{Born}} \left[1 + c_{\text{shell}} \frac{f_{\text{mix}} - 1}{|z_i|} \right], \quad f_{\text{mix}} = \frac{\sum_{j \in S} x_j f_{\text{solv},j}}{\sum_{j \in S} x_j}, \quad (17)$$

and replaces the radial factor in Equation 16 by

$$\frac{1 - 1/\epsilon_r}{d_i^{\text{out}}} + c_{\text{dielectric}} \left(1 - \frac{1}{\epsilon_{r,i}^{\text{local}}} \right) \left(\frac{1}{d_i^{\text{Born}}} - \frac{1}{d_i^{\text{out}}} \right), \quad \epsilon_{r,i}^{\text{local}} = 8. \quad (18)$$

The active parameter contract does not store these switches. If at least one ion declares a positive unique d_i^{Born} and at least one neutral component declares $f_{\text{solv}} \neq 1$, the Engine derives $c_{\text{shell}} = c_{\text{dielectric}} = 1$. Otherwise it derives zero for both. A declared ionic $d_i^{\text{Born}} = 0$ is not a zero physical radius: it is materialized as the same resolved ion diameter used by Debye–Hückel, $d_i^{\text{DH}} = 0.88\sigma_i$, which recovers the original-Born convention reported by Bülow et al. Explicit switch values remain accepted only for exact replay of older immutable packets. S is the explicit solvation pool: every component with a declared $f_{\text{solv},j}$. The primary Figiel calculations include all nine species in S ; the coupled CO_2 -exclusion calculation omits molecular CO_2 from both S and the salt-free neutral-permittivity pool. Thus “ CO_2 excluded” means both that CO_2

is not treated as a solvent-like contributor to f_{mix} and that it does not contribute to the salt-free neutral dielectric average; CO_2 remains a full reacting EOS component. No composition is silently renormalized outside the denominators shown above. Equation 18 is the Engine’s corrected nested radial charging expression, not the misprinted ordering in Figiel’s printed Eq. 7. The Engine differentiates Equations 16–18 with respect to temperature, density, composition, and active parameters.

The six fixed MEA-study Born diameters are 3.5332 Å (MEAH^+), 3.5411 Å (MEACOO^-), 3.0000 Å (HCO_3^-), 3.0000 Å (CO_3^{2-}), 1.2180 Å (H_3O^+), and 3.0811 Å (OH^-). They were held fixed in every structure comparison. Figiel et al. instead fitted aqueous infinite-dilution solvation data for H^+ , Li^+ , Na^+ , K^+ , VO_2^+ , V^{3+} , Cl^- , Br^- , I^- , and SO_4^{2-} ; their Table 2 diameters are respectively 1.218, 2.784, 3.445, 4.150, 2.930, 2.940, 4.100, 4.480, 4.985, and 4.953 Å [8]. Only the $\text{H}_3\text{O}^+/\text{H}^+$ value in the present bundle is an exact overlap. Figiel does not supply diameters for MEAH^+ , MEACOO^- , HCO_3^- , CO_3^{2-} , or OH^- ; substituting Na, halide, or sulfate values for those chemically different ions would not be source-consistent. A later diameter regression must therefore use ion solvation or other ion-specific evidence, not arbitrary cross-ion substitution.

The “ion-specific” α_i screen also needs qualification. Zuber et al.’s Table 2 directly gives 9.55 for H^+ and 13.96 for OH^- , which map exactly to the present H_3O^+ and OH^- records [22]. The present values 2.60 for MEAH^+ , 7.01 for MEACOO^- , and 7.89 for each carbonate species are project-declared analogs or fallbacks; Zuber did not fit those MEA-system ions. Consequently I-SSM+DS is an analog/fallback sensitivity, not a complete literature-matched ion-specific package.

The retained derivative audit independently perturbed one interior composition direction for Rules B and C. Analytic and centered finite-difference derivatives agree to relative errors of 2.21×10^{-11} and 1.95×10^{-12} , respectively; the exact inputs and step are retained in results/born-permittivity-study/component-permittivity-derivative-check.json.

11.2.2. Automatic activation and unique-diameter sensitivity

The current screen starts from the selected bundle in Tables 1–4: solvent-only mass-fraction permittivity, automatically activated SSM+DS, and $R5\ a_K = 2597.91$ K. It evaluates the complete parameter bundle against the same 12 retained sparse states for every perturbation. Each unique ion Born diameter was moved by $\pm 10\%$; $f_{\text{solv,MEA}}$ was tested at 1.0, 1.4, 1.5, and 1.6; and the $f_{\text{solv,MEA}} = 1.5$ case was crossed with each diameter movement. Pairwise statistics use only the exact positive targets shared by the perturbed and selected bundles, while failed states remain in the state table.

The selected bundle evaluates all 12 states. Its sparse pressure and species $\log_{10}$ RMSE are 0.4300 and 0.1246. Changing the H_3O^+ or OH^- diameter is numerically flat at this resolution. Individually, a 10% larger HCO_3^- diameter gives the clearest ion-radius direction, although two states fail and its pairwise improvements are only 0.0191 for pressure and 0.0033 for species. The MEA solvation-factor series is much more informative: 1.4, 1.5, and 1.6 all reduce both errors on all shared targets; 1.6 gives the smallest species RMSE and 1.4 the smallest pressure RMSE in that one-coordinate series.

The strongest fully evaluated joint direction is $f_{\text{solv,MEA}} = 1.5$ with a 10% larger HCO_3^- Born diameter. It retains all 12 states and changes pressure RMSE from 0.4300 to 0.3795 and species RMSE from 0.1246 to 0.0771. This is a screening direction, not a replacement parameter bundle; it is promoted below to the complete 205-state replay before any parameter is changed.

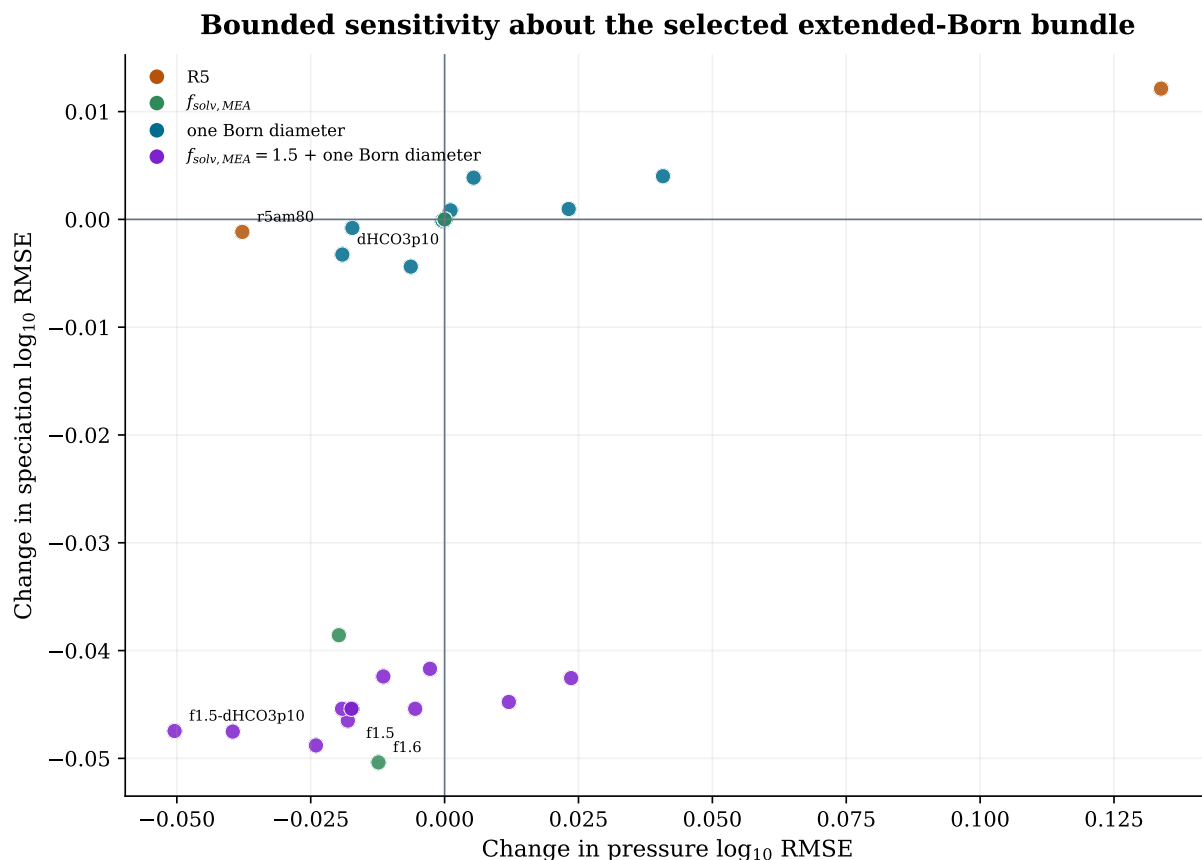


Figure 2: Pairwise common-target sensitivity about the selected extended-Born and R5-adjusted bundle. Negative changes improve agreement. The lower-left quadrant improves pressure and speciation simultaneously.

Exact row-level values, failures, pairwise comparisons, and the immutable receipt are retained under `results/born-permittivity-selected-bundle-joint-sensitivity-*`.

The complete replay changes the decision. On the 108 positive pressure targets evaluated by both bundles, the joint candidate reduces pressure $\log_{10}$ RMSE from 0.3424 to 0.2842 and median factor from 1.8364 to 1.5477. On all 131 common species targets, however, it increases speciation RMSE from 0.2373 to 0.2600. Numerical coverage changes only from 164 to 165 of 205 states. The direction is therefore useful for a later joint pressure, speciation, and calorimetry fit, but it does not replace the selected balanced bundle in Tables 1–4.

Full replay rejects the sparse candidate as a balanced replacement

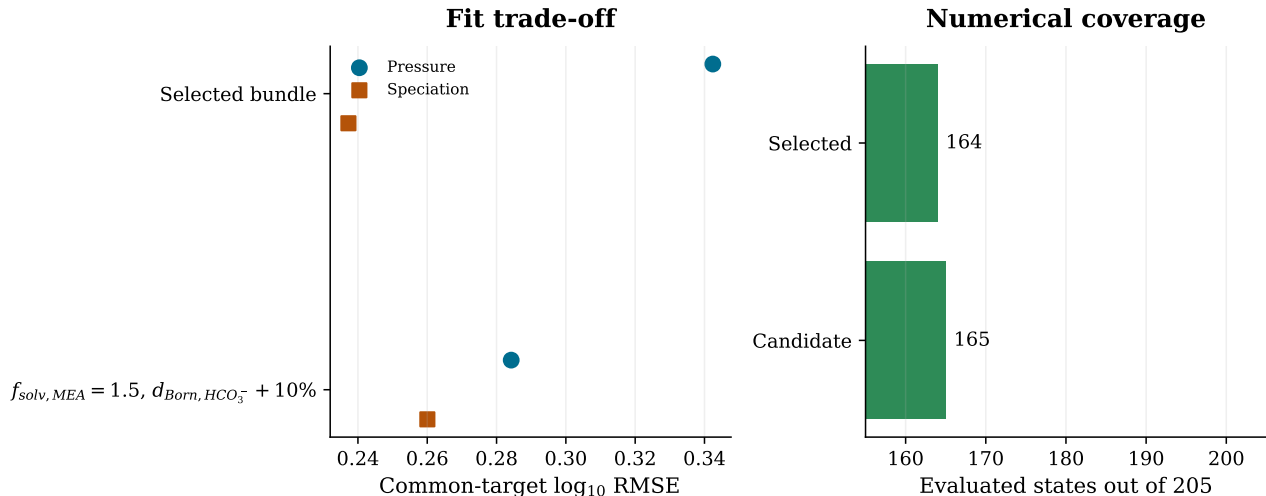


Figure 3: Complete 205-state comparison of the selected bundle and the best sparse Born/solvation direction. The pressure gain is real on identical targets, but it comes with worse full speciation.

11.2.3. Sparse structure screen

The first screen used 12 retained states: low, middle, and high loading at the two best-covered pressure temperatures (40 and 60 °C) and the two best-covered speciation temperatures (20 and 40 °C). All five permittivity rules were crossed with original Born and SSM+DS. The following table reports evaluated attempts and positive-target log₁₀ RMSE. The attempt count is a solver diagnostic, not an acceptance cutoff or scientific ranking coordinate.

Variant	States	ϵ_r range	p_{CO_2} RMSE	Pressure factor	Species RMSE
A original	12/12	57.13–79.29	0.2704	1.4008	0.0551
A SSM+DS	10/12	57.13–79.08	0.4910	2.7414	0.1358
B original	12/12	57.98–74.39	0.5017	2.6384	0.0728
B SSM+DS	9/12	57.57–74.28	0.5742	4.3025	0.1451
C original	12/12	43.03–63.78	0.9231	8.2675	0.0761
C SSM+DS	12/12	42.55–63.73	0.4020	2.4929	0.1317
D original	12/12	56.51–70.73	0.4645	2.8418	0.0514
D SSM+DS	11/12	56.50–69.66	0.5251	3.8518	0.1193
E original	11/12	34.63–60.22	1.3375	26.3051	0.0299
E SSM+DS	12/12	32.68–60.03	0.4669	2.2065	0.0960

SSM+DS does not improve every dielectric rule. It increases both errors for A, B, and D; it reduces the pressure RMSE by 0.5211 for C while increasing the species RMSE by 0.0555; and it reduces the E-rule pressure RMSE by 0.8705, but increases the species RMSE by 0.0660. Thus the apparent rescue of the Figiel package is primarily the corrected SSM+DS radial work, whereas the large baseline differences among A–E arise from the permittivity rule itself. This interaction rules out a single independent “Born correction” effect.

Because E–SSM+DS completed the sparse screen, the permitted follow-up tested universal 7.01 against Zuber-derived analog/fallback α_i , and $f_{\text{solv,MEA}} = 1.0, 1.4, 1.5, 1.6$. The 1.5 sparse case lost two states despite its smaller pressure RMSE; 1.4 and 1.6 completed all 12. An exact post-solve pool crossover at the 12 converged 1.6 compositions found $x_{\text{CO}_2} \leq 0.002918$, $|\Delta\epsilon_r| \leq 0.2663$ (0.813%), and $|\Delta f_{\text{mix}}| \leq 0.001249$ when CO₂ was excluded. That algebraic result was

used only as a cheap preliminary screen. The promoted comparison below then re-solved every state with molecular CO_2 removed from both pools so pressure, speciation, phase identity, and convergence were tested in the coupled system.

11.2.4. Complete retained-state comparison

Each promoted configuration is one complete nine-species parameter bundle. Every configuration attempted the same 161 active pressure observations from 313.15 to 393.15 K and the same 44 speciation states from 293.15 to 353.15 K; only the stated permittivity or Born coordinates differ among rows. This gives 1,640 configuration-state attempts in total. The Engine wheel used for that historical screen preserved the known-state result to numerical precision. Its parity timing was measured on the immediate predecessor candidate at Engine commit 355d950; the complete comparison used the final d8e02e4 immutable study wheel, which adds explicit neutral-pool exclusion, its tested derivatives, and preservation of final partial evidence at the unchanged bubble-iteration limit. The same solver path, tolerances, and starts were used. Those timings are retained only as historical reproducibility metadata; the reaction-synchronized scientific comparison below uses the current wheel and a three-CPU execution cap.

The Engine's fixed-pressure density anchor was measured separately at v1e_obs_0140. Eight single-threaded repetitions reduced median `Mixture.state` time from 58.98 to 14.76 ms and density iterations from 30 to 1 (4.00 times), with zero pressure or density difference. The present reactive equilibrium callable owns its density closure and does not expose this anchor, so the measurement did not alter the formulation campaign.

The separate 319-row cross-validation inventory remains non-executable: all 319 rows lack a residual scale and immutable data packet, and the 198 speciation rows also lack a fixed-pressure definition. The 123-state immutable packet used here is already executable and is not evidence that those broader admission decisions are resolved.

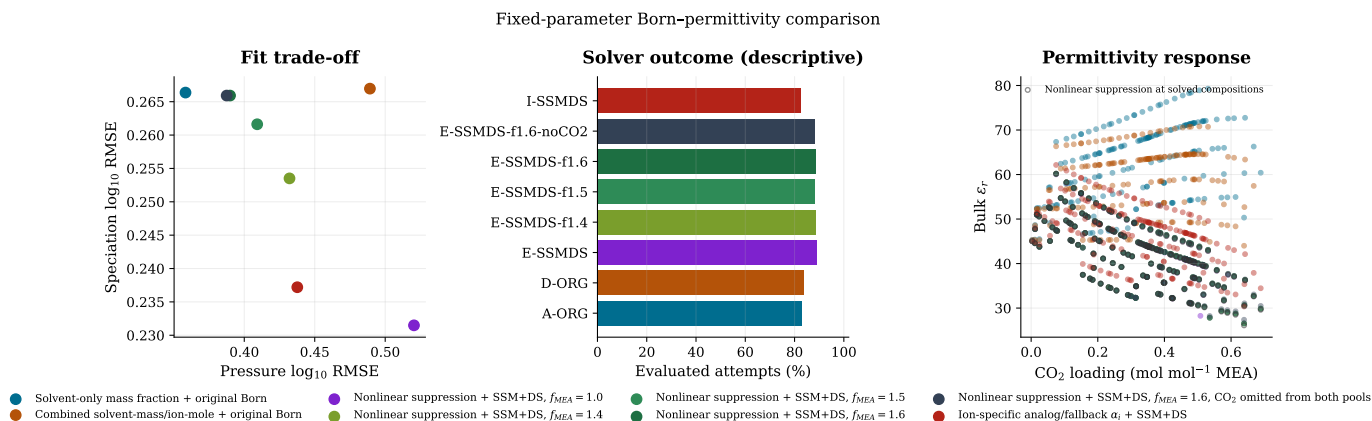


Figure 3: Fixed-parameter trade-off, descriptive evaluated-attempt fraction, and bulk relative permittivity for every promoted configuration. The open circles apply the nonlinear ionic-suppression equation to solvent-only-mass-fraction/original- Born solved compositions; they are a crossover, not independent loaded-MEA dielectric observations. Attempt fraction is not used to rank formulations.

Variant	$N_p/161$	p_{CO_2}	log RMSE	Factor	Bias	$N_x/44$	x log RMSE
A: solvent-only mass fraction + original Born	126/161		0.3583	1.9577	0.0956	44/44	0.2664
D: combined mixing + original Born	128/161		0.4892	2.5954	0.2763	44/44	0.2670
E: nonlinear suppression + SSM+DS, $f_{\text{MEA}} = 1.0$	139/161		0.5204	2.7461	0.2372	44/44	0.2315
E: nonlinear suppression + SSM+DS, $f_{\text{MEA}} = 1.4$	138/161		0.4321	2.2891	0.2177	44/44	0.2535
E: nonlinear suppression + SSM+DS, $f_{\text{MEA}} = 1.5$	138/161		0.4092	2.1997	0.2114	43/44	0.2616
E: nonlinear suppression + SSM+DS, $f_{\text{MEA}} = 1.6$	138/161		0.3899	2.1264	0.2074	44/44	0.2659
E: nonlinear suppression + SSM+DS, $f_{\text{MEA}} = 1.6$, no CO_2	137/161		0.3875	2.1090	0.2066	44/44	0.2659
Ion-specific suppression + SSM+DS	125/161		0.4376	2.3821	0.0710	44/44	0.2372

The table reports each complete bundle against both observation families. Its full-data metrics use every successful calculation for that bundle, so different numerical failures can change the rows entering a metric. A second, common-observation calculation therefore compares all eight bundles on the exact positive pressure-observation intersection ($n = 102$) and, separately, on the exact positive speciation-observation intersection ($n = 129$). The two intersections differ because pressure and liquid composition are different measured quantities, not because a different parameter bundle was used. On these common observations, solvent-only mass fraction with original Born has the smallest pressure log RMSE (0.3544; median factor 1.8607), while nonlinear suppression with SSM+DS and $f_{\text{MEA}} = 1.0$ has the smallest speciation log RMSE (0.2333). No row is selected from only one observation family. The complete common-observation table is retained in `results/born-permittivity-study/full-common-row-statistics.csv`.

Current reaction-synchronized result. The preceding eight-configuration screen is retained as model-structure history, but its observation requests embedded an older R4 correlation and therefore do not describe the present parameter bundle. The current comparison rebuilds every request from the displayed R1–R5 correlations and uses Engine commit 8438ce5f94a547189c91c4ec180a7782d60879d6. Each complete nine-species configuration was evaluated against the same 205 states. Metrics below use the exact intersection of 100 positive pressure targets and 123 positive speciation targets evaluated by all four configurations; coverage is reported separately, so no configuration benefits from silently dropping a difficult observation.

Complete configuration	States	p_{CO_2}	RMSE	Factor	Bias	Species RMSE	Species factor
Solvent-only mixing, original Born	171/205		0.3386	1.8652	0.0982	0.2734	1.0713
Solvent-only mixing, SSM+DS, literature R5	161/205		0.4930	2.4745	−0.4073	0.2676	1.2281
Selected: solvent-only mixing, SSM+DS, R5 $a_K = 2597.91$ K	164/205		0.3456	1.8672	−0.1532	0.2442	1.1466
Nonlinear suppression, SSM+DS, R5 $a_K + 80$ K	174/205		0.4417	2.3122	−0.0199	0.2440	1.0961

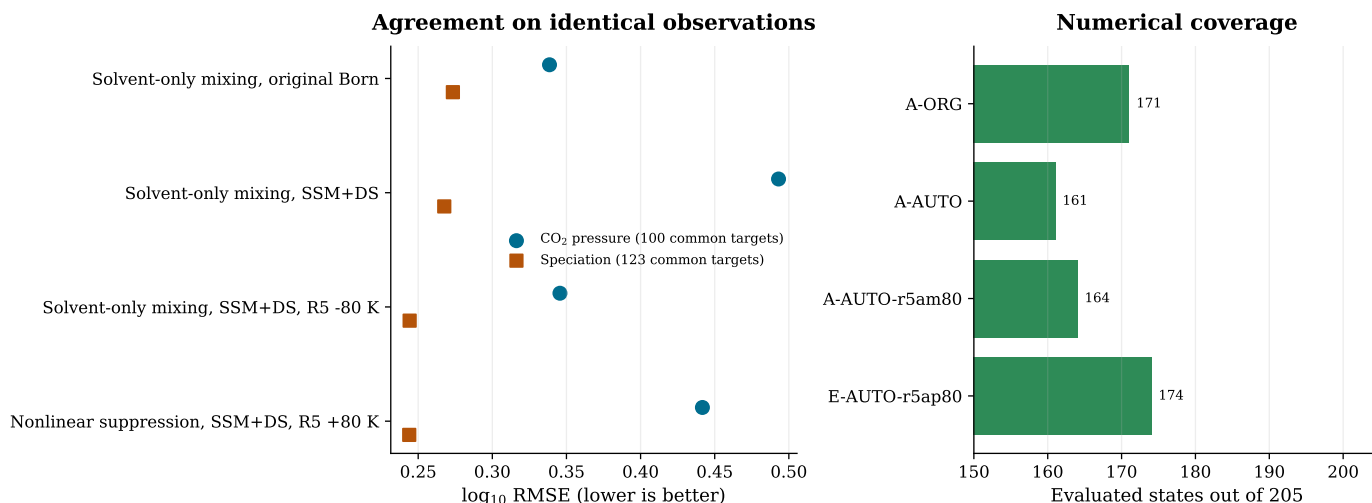


Figure 4: Current reaction-synchronized comparison. Pressure and speciation $\log_{10}$ RMSE use observations evaluated by all four configurations ($n_p = 100$, $n_x = 123$); the right panel reports all successful states out of 205 attempts. Lower RMSE is better.

No single row wins every quantity. Original Born gives the smallest common-row pressure RMSE. Nonlinear suppression with SSM+DS and R5 $a_K + 80$ K gives the largest evaluated-state count, nearly zero mean pressure bias, and the smallest species RMSE, but its pressure scatter remains larger. The best tested balance and the selected active bundle is solvent-only mixing with SSM+DS and R5 $a_K = 2597.91$ K: relative to original Born, its pressure RMSE is only 0.0070 larger while its species RMSE is 0.0292 smaller. Relative to the unadjusted SSM+DS bundle, it reduces pressure RMSE by 0.1474 and species RMSE by 0.0233.

The opposite useful R5 directions under the two permittivity rules are the central result. R5 is not an independent correction that can be transferred unchanged between Born/permittivity formulations: reaction speciation changes liquid composition and CO₂ activity, which changes bubble pressure, while the electrostatic formulation changes the activities entering R5. The fixed-R5 comparison therefore diagnosed coupling rather than proving that original Born is intrinsically superior. A defensible next fit must estimate R5 together with the uncertain ion Born diameters and neutral solvation factors under one chosen permittivity formulation, using both pressure and speciation residuals.

The four current calculations preserve 34, 44, 41, and 31 failed states, respectively. The nonlinear-suppression row improves coverage but does not eliminate the high-temperature pressure failures. Those states remain solver evidence and must be recovered before the formulation comparison is considered complete. Exact values are retained in the Born-permittivity study results directory: `current-fast-common-comparison.csv`, with row-level states and targets in the `corrected-reaction-full` and `fig1-r5-positive-full` result families.

Figure 5 shows that every configuration shares the same principal pressure pattern: negative low-loading residuals, positive middle-loading residuals, and a temperature-dependent high-loading return. The speciation statistic is dominated by underprediction of HCO_3^- ; the other reported quantities cluster near zero residual. The grouped heat map makes the temperature, source, and species dependence explicit.

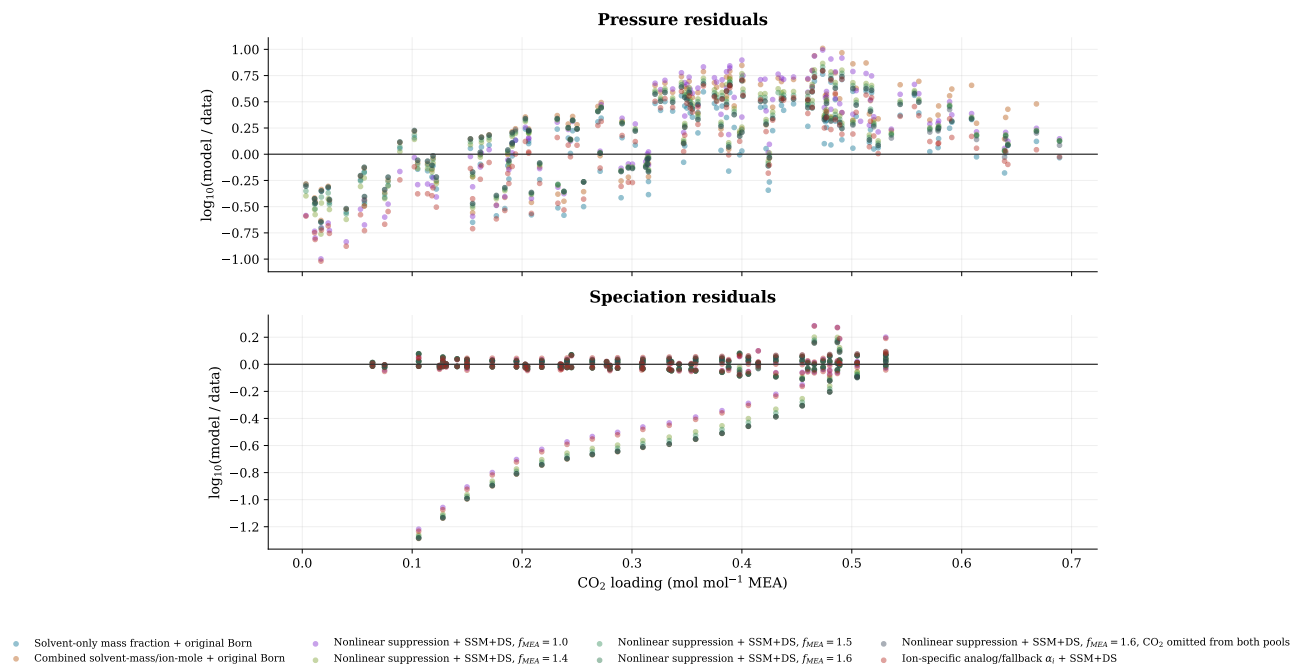


Figure 5: All positive observed–predicted pressure and speciation residuals. Every point is a direct retained Engine evaluation; no curve or failed state has been interpolated.

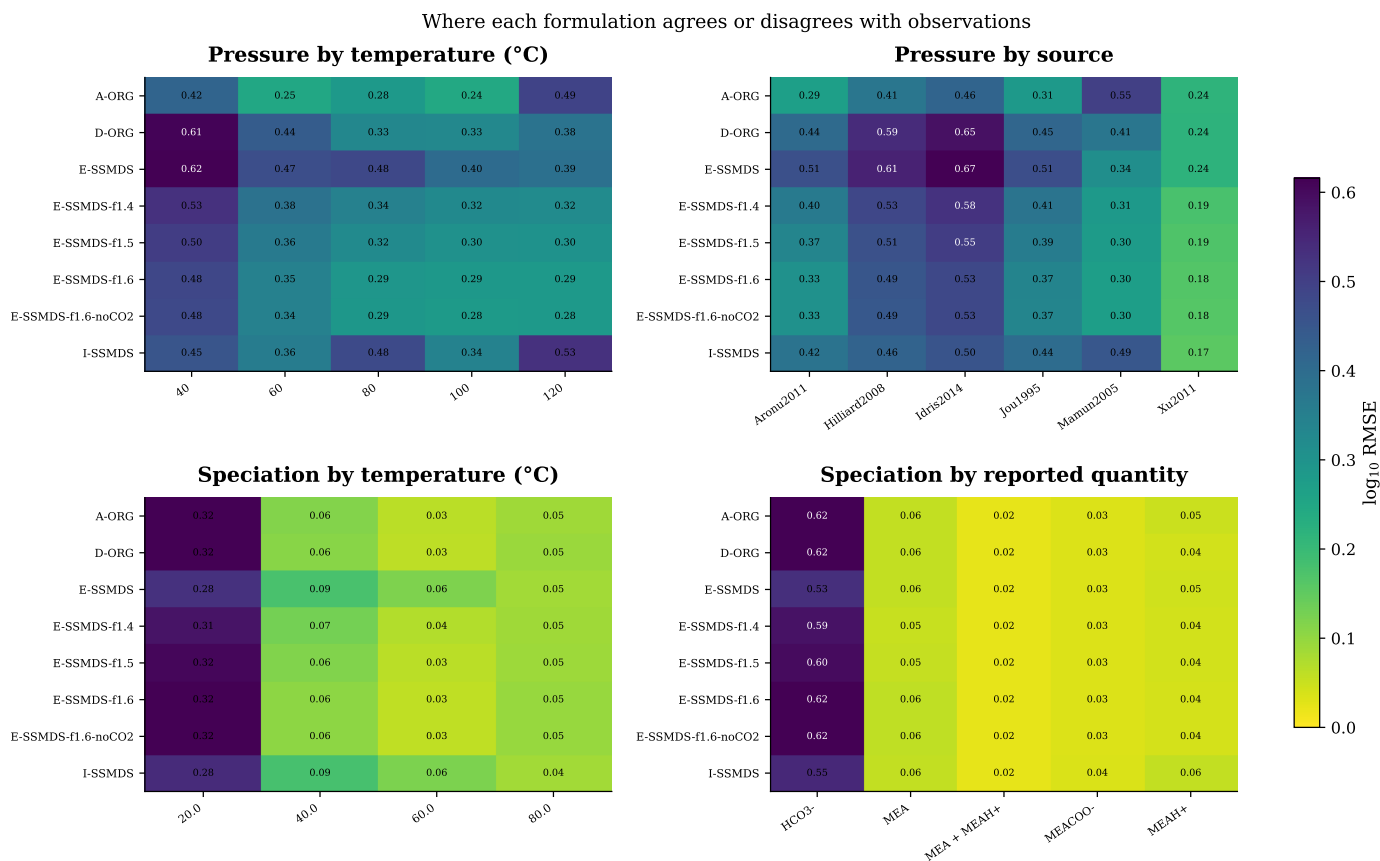


Figure 6: Log₁₀ RMSE grouped by pressure temperature and source, and by speciation temperature and reported quantity. Missing groups remain missing rather than being imputed.

Every evaluated liquid state is on an accepted liquid mechanical branch, and every incipient vapor is on its accepted

vapor branch. Maximum absolute charge closure is below 2.4×10^{-17} , material closure below 2.8×10^{-13} , reaction-affinity closure below 1.3×10^{-11} , and relative pressure closure below 6.4×10^{-10} . Figure 7 retains the density range, closure values, and all 220 non-evaluatable attempts. Of those, 181 reach the existing bubble-pressure Newton iteration limit, 38 produce a finite phase candidate that does not pass the complete equilibrium certificate, and one stops in the inner IPOPT solve. The complete row-by-row table records source, temperature, loading, continuation and start availability, and stopping point. The former label “request-policy validation” was a reporting defect: an empty policy record hid the native “bubble Newton iteration limit reached” result. Engine commit d8e02e4 preserves the last partial state and policy so that the actual failure is now retained. These are execution boundaries, not evidence that a physical formulation is false.

Source	Temperature (°C)	Bubble iteration	Certificate	Inner solve
Aronu 2011	40, 60, 80	0	14	0
Böttinger 2008	20	0	1	0
Hilliard 2008	40	0	13	1
Idris 2014	40	0	1	0
Jou 1995	40, 60, 80, 120	42	5	0
Mamun 2005	120	70	1	0
Xu 2011	100, 120	69	3	0
Total		181	38	1

These counts are formulation–state attempts over 60 distinct observations. All failures had a continuation value. Of the 181 bubble-iteration stops, 165 used one pressure start without a liquid start and 16 also had a liquid start. Of the 39 later stops, 38 used one pressure start and one certification failure had no pressure start. The exact loading and per-variant record remains in `results/born-permittivity-study/full-failure-characterization.csv`.

All 38 certificate failures compiled, completed the inner solve, passed the EOS-domain check, and returned a locally evaluatable candidate, but no candidate passed the full closure and local-stability certificate. Thirty-seven are reported as numerical-convergence failures; one E-SSM+DS, $f_{\text{MEA}} = 1.5$ attempt at 20 °C is a physical-rejection classification with material closure 0.0243, reaction-affinity closure 1.893, KKT closure 0.0125, and failed local stability. The same observation evaluates under the other seven configurations. The single IPOPT failure likewise occurs for one configuration of a row evaluated by most alternatives. These facts make all 39 candidate/inner-solve failures branch-and-initialization robustness evidence, not proof that the observed state or formulation is physically impossible.

The 181 bubble failures are concentrated at high temperature: 182 of all 220 non-evaluatable configuration–state attempts occur at 120 °C, with the remainder at 20, 40, 60, 80, and 100 °C. Increasing the outer iteration cap from 18 to 60 and sweeping pressure starts from 10 kPa to 5 MPa did not recover the representative 120 °C row; iterates alternated between gas-like and dense-liquid roots. The required Engine experiment is therefore a branch-aware coupled pressure KKT formulation, not looser tolerances or more iterations. Every one of these 181 rows is retained as an acceptance case: the structural comparison must be rerun after that solver change, and no present attempt count is used to rank formulations.

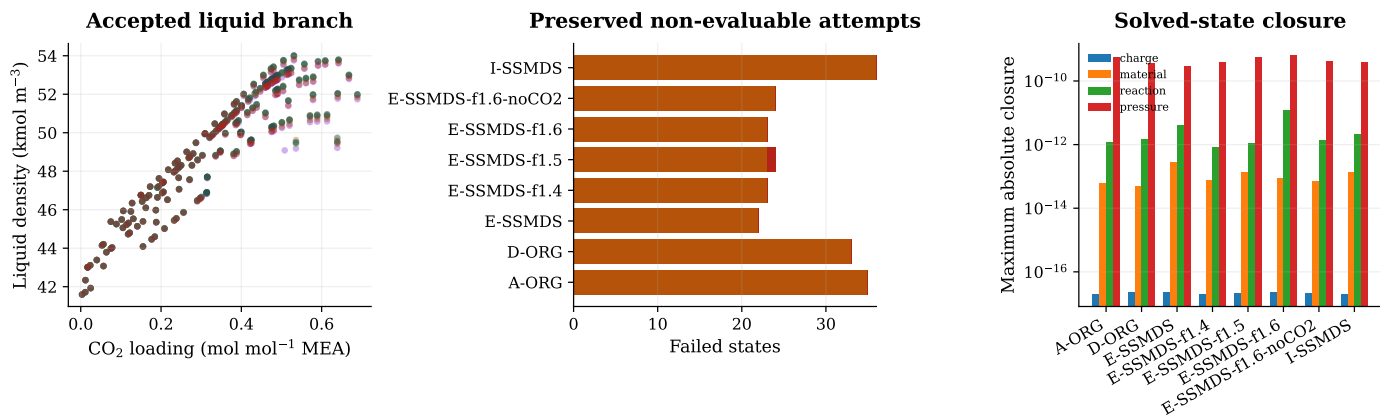


Figure 7: Accepted liquid density, preserved failure classes, and maximum charge, material, reaction-affinity, and pressure closure for every promoted configuration.

At the common 40 °C, loading 0.389 state, A-original has $(a^{\text{Born}}, a^{\text{DH}}, a^{\text{res}})/(RT) = (-13.8158, -0.0577, -22.4172)$; E-SSM+DS with $f_{\text{MEA}} = 1.6$ gives $(-12.7242, -0.0989, -21.3758)$. The separated axes in Figure 8 expose the otherwise hidden Debye-Hückel change.

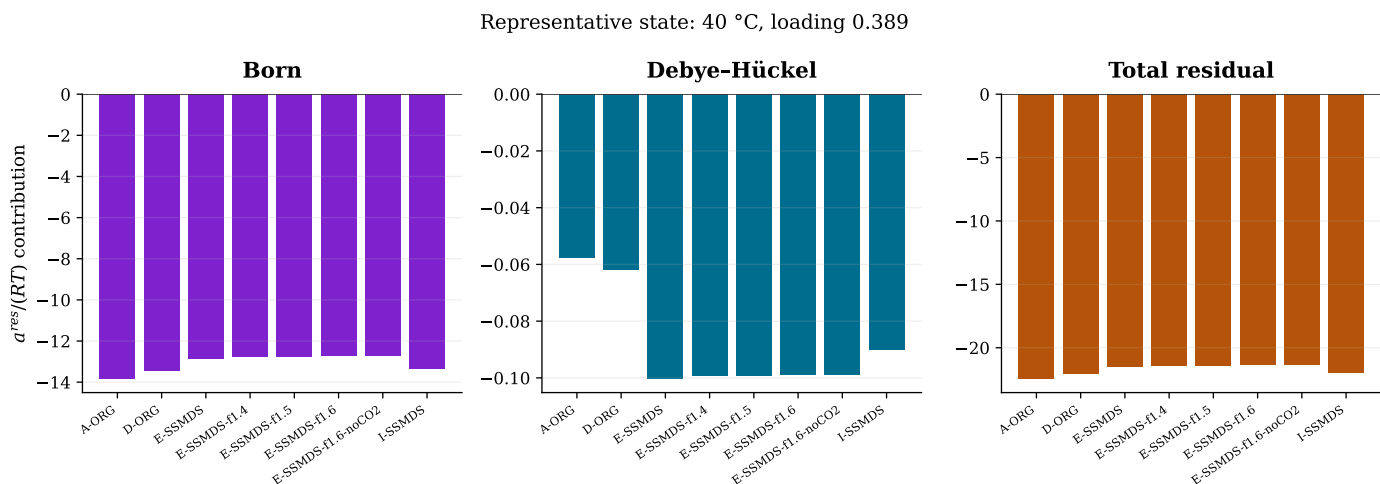


Figure 8: Born, Debye-Hückel, and total residual Helmholtz contributions at one common representative state. Each panel has its own ordinate scale.

The retained dielectric folder contains no admitted static-permittivity data for loaded aqueous MEA. The plotted Figiel relation is therefore a literature-form crossover evaluated at this study's solved compositions, not independent MEA validation. Within that limit, E predicts 26.54–60.19, the Zuber analog/fallback suppression screen predicts 30.36–62.15, D predicts 44.00–70.80, and A predicts 45.32–79.29 across successful states. Coupled molecular- CO₂ exclusion from both f_{mix} and the neutral dielectric pool changes the universal-1.6 pressure log RMSE from 0.389898 to 0.387514, changes species log RMSE by less than 2×10^{-7} , and has negligible effect on the present common-observation residuals.

Current interpretation: populated unique Born diameters and non-unit neutral solvation factors activate SSM+DS. With the unadjusted R5 correlation, that formulation worsens pressure. A bounded R5 movement nearly recovers the original-Born pressure result and improves speciation, so the discrepancy is a coupled reaction/electrostatic calibration problem rather than evidence for turning SSM+DS off. Solvent-only mixing with SSM+DS and R5 $a_K = 2597.91$ K is the selected active configuration; nonlinear suppression with R5 $a_K + 80$ K is the coverage and pressure-bias leader. Neither settles the permittivity model without loaded-MEA dielectric observations and a consistent Born-diameter/R5 fit. Explicit

CO₂ exclusion remains immaterial at the present compositions.

12. Current conclusions and next experiments

The selected physical structure includes reciprocal induced CO₂-H₂O association and excludes CO₂ self-association. The active foundation uses corrected R1–R3, solvent-only mass-fraction dielectric mixing, automatically activated SSM+DS, and $k_{\text{CO}_2, \text{MEA}} = 0$. The following steps provide the shortest discriminating path. Items below are proposed work, not retained results.

1. Preserve the certified loading/temperature continuation replay that recovers all 161 pressure observations without changing parameters or tolerances. Reduce the remaining three optional speciation display-grid failures with the same branch-preserving method, and use those cases as acceptance evidence for the upstream unified pressure–equilibrium KKT owner.
2. Complete the coupled bubble-pressure ablation with the active parameter file: nine versus six liquid species, crossed with ePC-SAFT versus ideal vapor. The retained comparison has already established the exact six-species reaction projection, ideal-liquid behavior, fixed- T , P liquid behavior, and the failure of the neutral apparent-flash shortcut. Report state evaluability, pressure and species metrics, charge/balance closure, and changes by pressure range and temperature.
3. Freeze calibration and held-out source/temperature partitions, then replay the sensitivity screen with the active parameter packet. Limit the first pass to R4/R5, the bicarbonate and carbonate Born diameters, the three Class 2 nonzero water–trace-ion corrections, and the seven named Class 2 zero pairs. Do not run a broad fit until the recovered-state sensitivities show which coordinates are identifiable from both pressure and speciation.
4. Seek simpler identifying evidence for the weak blocks: protonated-MEA salt and bicarbonate activity, osmotic, density, or solvation data. Also admit independent physical-solubility evidence before expanding the molecular model. Jiru et al. measure physical CO₂ solubility in MEA–water by the N₂O analogy over 298.15–323.15 K [13]; Hartono et al. report N₂O solubility in unloaded and loaded MEA [10]. Wong et al. directly measure dissolved molecular CO₂ and Henry behavior [21], but every retained row contains 0.517 mol kg^{−1} NaClO₄, which the nine-species model omits; use those rows as an external challenge unless the salt is represented explicitly.
5. Use the completed 113-row direct total-enthalpy replay to identify the smallest reaction-temperature correction supported simultaneously by pressure, speciation, and calorimetry. Keep the 66 Kim–Svendsen 40/80 °C rows as calibration, the 20 rows at 120 °C as a temperature holdout, and the 27 Kim 2014 rows as model-selection comparison evidence. Before fitting, test the declared zero-vapor/ideal-feed reconstruction and the first-row loading assumption. Do not copy the eNRTL block weight or numerical reaction values.
6. Acquire or digitize independent static-permittivity observations for loaded aqueous MEA. The completed fixed-parameter comparison cannot select between the A and E dielectric responses from pressure and speciation alone, and the Figiel relation is not an independent MEA measurement. Keep solvent-only mass-fraction mixing as the active dielectric rule until that evidence exists.
7. Continue Figiel’s staged identification instead of fitting every radius and solvent factor at once. Keep the literature-anchored water factor at 1.5, omit molecular CO₂ from the solvent pool because its retained crossover was immaterial, and profile $f_{\text{solv}, \text{MEA}}$ jointly with each uncertain ion Born diameter (with the two trace-carbonate diameters also tested as a pair). Every grid member must use the same pressure and speciation rows. Promote only coordinates whose pressure and species sensitivities are both non-flat and not mutually compensating; then perform one bounded joint iteration and a held-out replay. This is the shortest MEA analogue of Figiel’s diameter–factor– k_{ij} fitting sequence and directly tests practical identifiability.
8. Only after these identifying experiments, run a bounded joint optimization over the supported coordinates. With R1–R5, the ion block, and independently accepted $k_{\text{CO}_2, \text{H}_2\text{O}}$ fixed, compare only physical dissolved-CO₂ alternatives that the new data can distinguish: a CO₂–MEA association edge, a physically identified solvation/activity contribution, and any richer binary correction justified by the temperature and composition coverage. Do not combine

coordinates that the admitted observations cannot identify.

9. Require simultaneous comparative improvement in pressure, speciation, and convergence coverage, followed by held-out temperatures and sources before issuing an immutable packet. No unapproved numerical threshold is an acceptance gate.

unknown: whether ideal vapor, reduced chemistry, or neither materially changes the coupled bubble-pressure residual has not been tested with the active bundle.

Replace the current numerical foundation in place when a better MEA-owned regression or validation result passes the stated comparison gates. Every replacement must bind the Engine wheel and hash, parameter identities and evidence status, admitted rows and hashes, optimizer diagnostics, validation summaries, and the transfer decision. This PDF is the live calculation and interpretation record for the active bundle, including the reported fit statistics and state counts.

12.1. Independent review record

An independent read-only review on 2 September 2026 assessed commit fb6f948. It verified the 161-pressure plus 44-speciation scope, the coupled CO₂-pool exclusion, normalized f_{mix} , separate source hashes, Rules B/C derivative checks, preserved failures, recomputed summary metrics, and the synchronized 27-page render. The verdict was *Review Pass*: no material scientific, numerical, provenance, or layout finding remained. The approved use is fixed-parameter model-structure selection and failure characterization. It does not promote a new active bundle, resolve the separate 319-row admission decisions, or supply missing independent loaded-MEA static-permittivity observations.

13. Provenance

The base numerical reconstruction uses the PR #134 file:

`validation/campaigns/2026-mea-parameter-estimation/analysis/joint-fit/parameters.json`

at commit 38e91823b6d4f26c1d549f07aaef24a089d8e16d. The selected live mapping is:

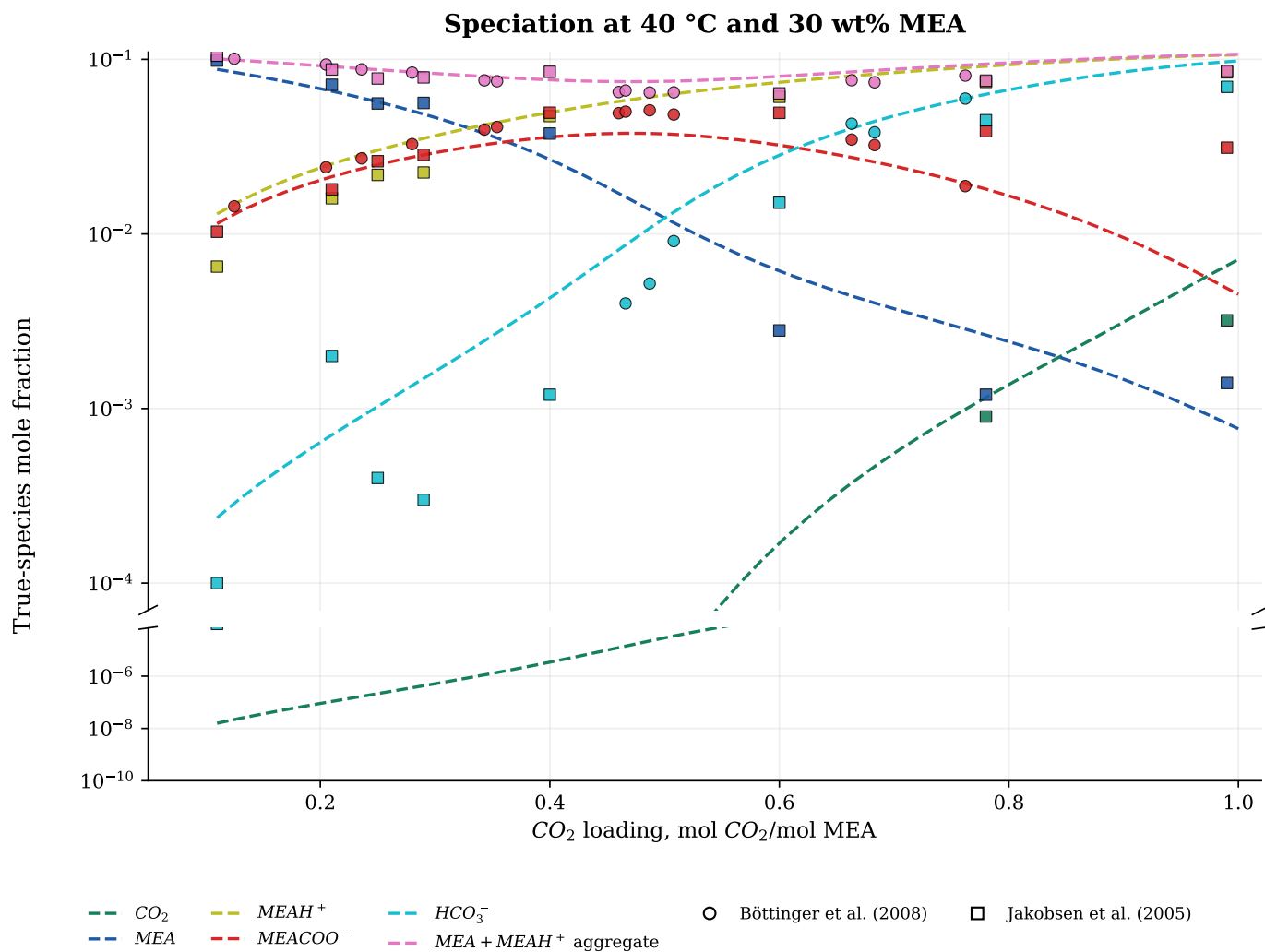
`results/selected-current-best-parameters.json`

Its SHA-256 is the parameter identity reported in Section 2. The comparison result records both the base and selected fingerprints.

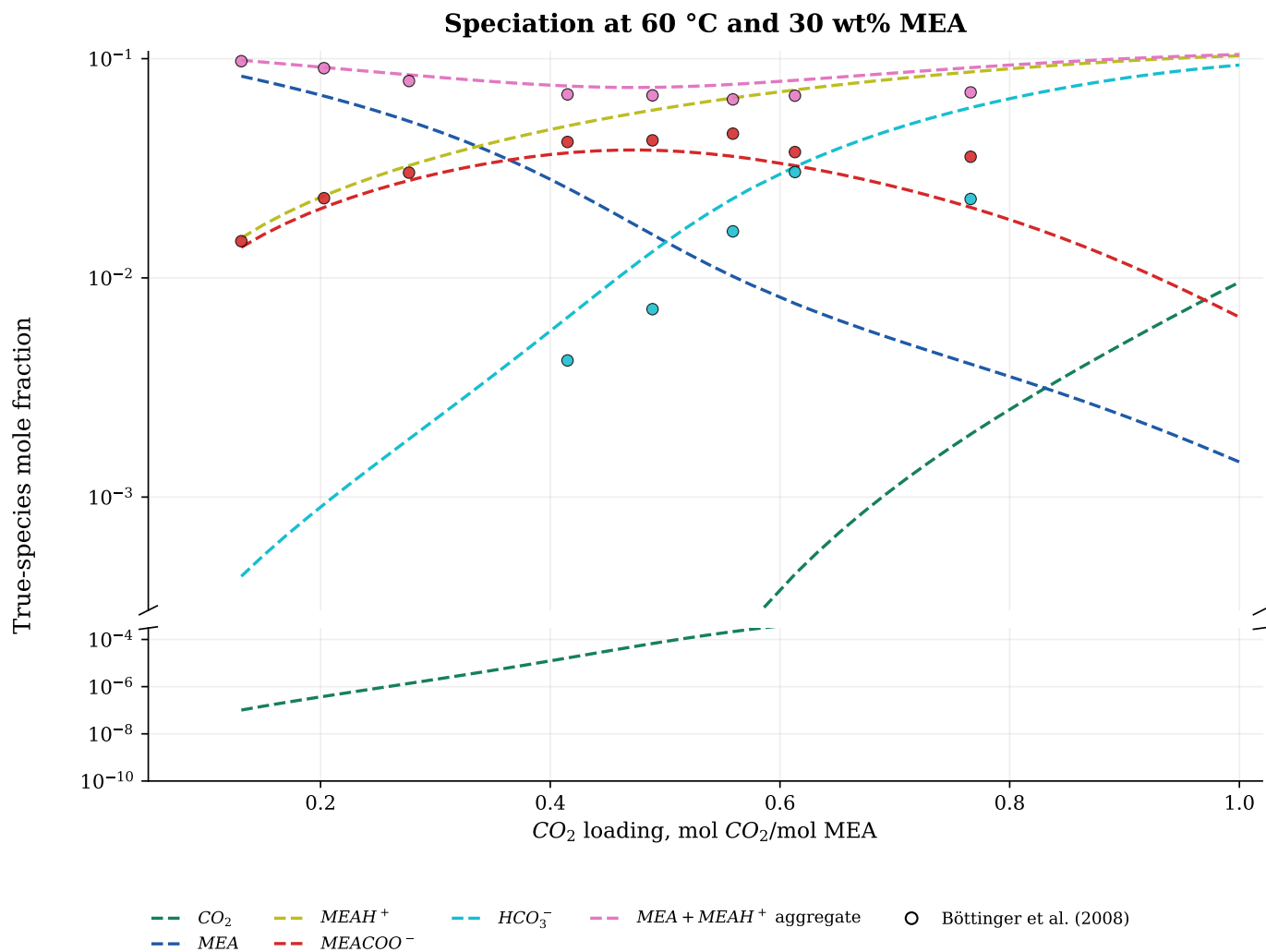
Source	Role
ePC-SAFT Issue #80	Upstream full regression and immutable-packet campaign.
ePC-SAFT Issue #119	Bounded refinement scope and current foundation reconciliation.
ePC-SAFT PR #134	Closed, unmerged blocker-evidence change. It produced no parameter bundle; this notebook reconstructs only its frozen foundation vector.
Local component source audit	docs/ePC-SAFT/full-component-parameter-source-audit.md; species-level lineage and evidence gaps.
Local model-hierarchy review	docs/ePC-SAFT/amine-epcsaft-model-hierarchy-literature-review.md; selected nine-species structure and induced-association decision.
Retained parameter evidence	analyses/reactive_epcsaft_parameter_evidence/; supported-negative and diagnostic evidence.
Retained local-speciation fit	Git commit 897bb6e; the May 2026 8-state/22-residual principal-ion fit retained as Class 3 evidence awaiting promotion checks.
Reaction exporter defect	validation/campaigns/2026-mea-parameter-estimation/scripts/build_fit_input.py at Engine commit 38e91823b6d4f26c1d549f07aaef24a089d8e16d.
Corrected start comparison	results/parameter-start-comparison.csv; full corrected-foundation and Böttinger-R4 comparison statistics.
Bounded sensitivity screen	results/parameter-sensitivity-screen.csv; common-row CO ₂ -MEA and CO ₂ -water perturbation results used in Section 7.
Quick direct endpoint screen	results/quick-endpoint-perturbation-screen.csv and its JSON receipt; full reactive bubble-point endpoints and the bounded SciPy reaction-root spot check.
Best-in-slot campaign	results/best-in-slot-campaign/; centered sensitivity, R4 pivot-slope grids, full validations, boundary diagnostics, exact receipts, and the selected candidate's row-level Engine evaluations.
Born-permittivity structure study	results/born-permittivity-study/; exact sparse and complete state/target tables, failure characterization, grouped statistics, Engine parity and density-anchor timings, deterministic formulation construction, and hashes. Figures 3–8 read these retained values without rerunning the Engine.
Current Engine wheel	data/input/engine/epcsaft-0.2.0.dev0-cp313-cp313-linux_x86_64.whl; Engine commit 8438ce5f94a547189c91c4ec180a7782d60879d6 and wheel SHA-256 40fba7cfb9c8414152f3e49636c49ae2e3f7099e30040d54d464ccb38355f805.
Selected live mapping	results/selected-current-best-parameters.json; solvent-only mass-fraction mixing, automatic SSM+DS activation, the selected Class 3 R4-CO ₂ fit, and R5 $a_K = 2597.91$ K.
Trace-species abundance	figures/speciation/output/speciation-model.csv; retained 44-state calculated values summarized in Section 10.
Current IDAES comparison	Official public MEA solvent and MEA vapor property files; used only to define the proposed structural ablation.
Matched species comparison	../enrtl_six_species_ideal_comparison/notebook.qmd and ../enrtl_six_species_ideal_comparison/notebook.pdf; exact ideal projection, fixed- T , P Engine/SciPy comparison, corrected apparent-pressure diagnostic, parameter inventory, sensitivity survey, UQ checkpoints, and figures.

A. Additional-temperature speciation replays

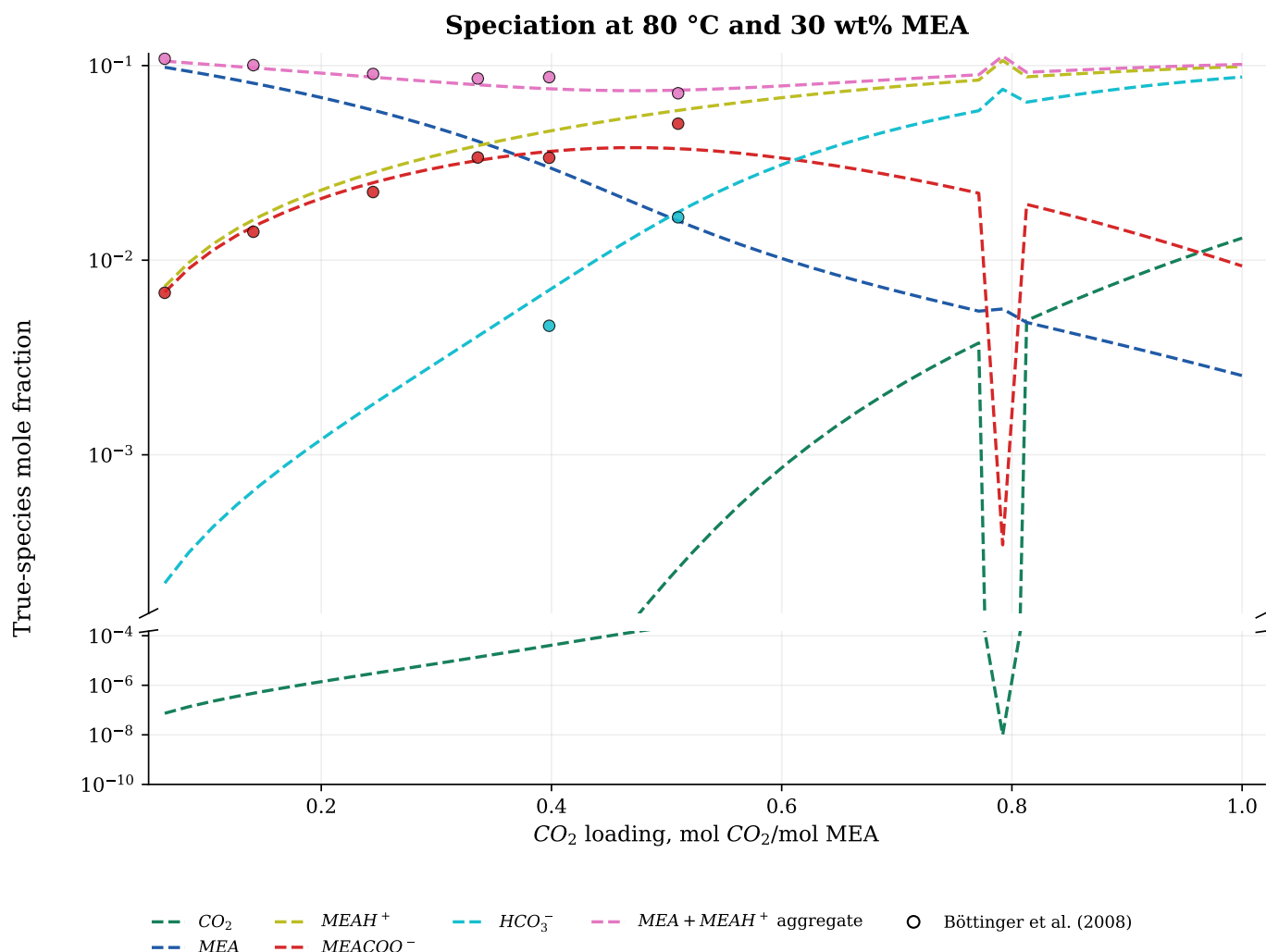
The following panels apply the same species selection, source encoding, reported-zero omission, split-axis rule, parameter bundle, and pinned Engine route as Figure 1. They are supplementary temperature views rather than additional fits. At 40 °C, 74 positive plotted values cover 21 states and observed loading 0.11–0.99; at 60 °C, 21 values cover eight states and observed loading 0.131–0.766; at 80 °C, 14 values cover six states and observed loading 0.064–0.51. Every model grid continues to loading 1.0, and all 46 states evaluated at each temperature. Dashed lines connect those direct calculations.



Appendix Figure A1. Principal-species replay at 40 °C and 30 wt% MEA. Symbols retain every displayed Böttinger and Jakobsen observation; dashed lines are direct Engine calculations.



Appendix Figure A2. Principal-species replay at 60 °C and 30 wt% MEA. Symbols retain every displayed Böttinger observation; dashed lines are direct Engine calculations.



Appendix Figure A3. Principal-species replay at 80 °C and 30 wt% MEA. Symbols retain every displayed Böttinger observation; dashed lines are direct Engine calculations.

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